# Supplementary Material for "FundusShift and LOTUS: Benchmarking and Improving Fundus Disease Classification at Unseen Sources"

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This supplement gives every table behind the main paper, plus analyses that did not fit there. All tables are generated from the saved results in the repository (`https://doi.org/10.5281/zenodo.23979471`) by `paper/latex/make_assets.py`, `make_supp.py` and `make_supp_extra.py`; LOTUS is the final method, and the *initial design* is LOTUS with the between-site projection of the main paper (Sec. 5); numbers are means over seeds unless stated. Abbreviations: bAcc, balanced accuracy; Eq-src, mean of per-source bAcc; R/T/F, raw, temperature scaling (TS), focal temperature scaling (FTS); LP, linear probe; LAC, least-ambiguous conformal sets; SGR, selective guaranteed risk; FR, disease images answered normal, over all disease images; G→N, glaucoma images accepted and predicted normal, over all glaucoma images.

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## S1 Data

### S1.1 Sources, label scope and licences

Table S1 lists every source with its class counts per split. "Scope" records which diseases a source graded: in DR-only sources a normal label means "no DR", and glaucoma was not assessed.

Table S1: Sources. Counts are N/D/G. Development sources contribute train, validation and calibration; test sources and Chákṣu contribute only the evaluation split.

| Source | Country | Scope | Licence | Train | Validation | Calibration | Test or OOD |
|---|---|---|---|---|---|---|---|
| Bangladesh Eye | Bangladesh | full | CC BY 4.0 | 677/1200/925 | 85/150/116 | 84/150/115 | 0/0/0 |
| OIA-DDR | China (147 hospitals, 23 pro | DR only | CC BY-NC-SA 4.0 | 5012/5004/0 | 627/626/0 | 626/625/0 | 0/0/0 |
| DeepDRiD | China | DR only | CC BY-SA 4.0 | 732/868/0 | 94/110/0 | 88/108/0 | 0/0/0 |
| GRAPE | China | full | CC0 | 0/0/470 | 0/0/58 | 0/0/58 | 0/0/0 |
| IDRiD | India | DR only | CC BY 4.0 | 133/277/0 | 17/35/0 | 17/35/0 | 0/0/0 |
| Jichi | Japan | DR only | CC BY 4.0 | 5028/2504/0 | 635/308/0 | 632/308/0 | 0/0/0 |
| SUSTech-SYSU | China | DR only | CC BY 4.0 | 505/470/0 | 63/59/0 | 63/59/0 | 0/0/0 |
| AFIO | Pakistan (Armed Forces In-sti | full | CC BY 4.0 | – | – | – | 67/18/44 |
| FIVES | China | full | CC BY 4.0 | – | – | – | 193/200/193 |
| HRF | Germany / Czech Republic | full | CC BY 4.0 | – | – | – | 15/15/15 |

Table S1: (continued)

| Source | Country | Scope | Licence | Train | Validation | Calibration | Test or OOD |
|---|---|---|---|---|---|---|---|
| HYGD | Israel | full | ODC-By 1.0 | – | – | – | 0/0/548 |
| PALM | China (Zhongshan Ophthalmic | full | CC0 | – | – | – | 161/0/0 |
| PAPILA | Spain | full | GPL-3.0 | – | – | – | 333/0/87 |
| Paraguay | Paraguay | DR only | CC BY 4.0 | – | – | – | 708/723/0 |
| Chákṣu | India | full | CC BY 4.0 | – | – | – | 1156/0/187 |

*Reading*. The development side holds 1,395 glaucoma training images, all from Bangladesh Eye (925) and GRAPE (470), whereas DR and normal images come from the six sources other than GRAPE. Five of seven development sources graded only DR. On the test side, only Paraguay is DR-only, HYGD is G-only and PALM is N-only, so per-source bAcc on those two reduces to one recall. These facts explain much of the glaucoma behaviour in the main paper.

## S1.2 Source hierarchy, patients and label scope

FundusShift holds out whole public *collections* (sources), not hospitals. Table S2 gives the institution or country behind each collection, the number of patients where the source releases identifiers, the classes it contributes and what its "normal" label means. All but one collection come from a single institution or study; OIA-DDR pools 147 hospitals, so it is one development source with internal variety, not 147 held-out units. Each collection was also assigned an acquisition family (institution), and the audit of Sect. S1.5 confirms that no family spans the development and test sides; RFMiD 2.0, from the same institution as IDRiD, is used only for duplicate checks.

**Table S2.** Source hierarchy and label scope. Patients: distinct patient identifiers among used images (n/a: not released; for Chákṣu only the 60 Bosch-camera images carry identifiers, so its other images form leakage groups by duplicate detection alone and are treated as independent in bootstrap intervals). Normal means: what the source's normal label certifies.

| Role | Source | Institution / country | Images | Patients | Classes | Normal means |
|---|---|---|---|---|---|---|
| Dev | Bangladesh Eye | Bangladesh | 3,502 | n/a | N/D/G | no graded disease |
| Dev | OIA-DDR | China (147 hospitals, 23 provinces) | 12,520 | n/a | N/D | no DR (DR-only grading) |
| Dev | DeepDRiD | China | 2,000 | 500 | N/D | no DR (DR-only grading) |
| Dev | GRAPE | China | 586 | 133 | G | no graded disease |
| Dev | IDRiD | India | 514 | n/a | N/D | no DR (DR-only grading) |
| Dev | Jichi | Japan | 9,415 | 2,726 | N/D | no DR (DR-only grading) |
| Dev | SUSTech-SYSU | China | 1,219 | n/a | N/D | no DR (DR-only grading) |
| Test | AFIO | Pakistan (Armed Forces Institute of Ophthalmology, Rawalpindi) | 129 | 65 | N/D/G | no graded disease |
| Test | FIVES | China | 586 | n/a | N/D/G | no graded disease |
| Test | HRF | Germany / Czech Republic | 45 | n/a | N/D/G | no graded disease |
| Test | HYGD | Israel | 548 | 186 | G | no graded disease |
| Test | PALM | China (Zhongshan Ophthalmic Center) | 161 | n/a | N | no graded disease |
| Test | PAPILA | Spain | 420 | 210 | N/G | no graded disease |
| Test | Paraguay | Paraguay | 1,431 | n/a | N/D | no DR (DR-only grading) |
| OOD | Chákṣu | India | 1,343 | 30 (60 img.) | N/G | no graded disease |

*Reading.* Five of seven development sources and one test source (Paraguay) graded DR only, so their normal images were never assessed for glaucoma. Only four test sources assessed their normal images for glaucoma (AFIO, FIVES, HRF, which graded all three classes, and PAPILA, which graded N and G); Table S16 reports test bAcc restricted to them, and the label-scope experiment of Sect. S8 tests treating DR-only normals as possibly glaucomatous. Because glaucoma appears in only two development sources, glaucoma results measure transfer from those two sites, and the rotations (Table S29) show how they change with more glaucoma sources.

### S1.3 Image quality by source

**Table S3.** Image properties of used images by source (medians): original size, field-of-view fraction, brightness (mean luminance in the field), RMS contrast, Laplacian sharpness, and the share flagged as very dark, overexposed, low field of view or blurry for its source.

| Source | Size | FOV | Brightness | Contrast | Sharpness | Flagged |
|---|---|---|---|---|---|---|
| Bangladesh Eye | 2,004×1,690 | .838 | 111.2 | 22.5 | 65.0 | 0.8% |
| OIA-DDR | 2,592×1,824 | .535 | 83.4 | 20.8 | 61.6 | 0.8% |
| DeepDRiD | 1,736×1,824 | .682 | 86.2 | 29.4 | 38.2 | 2.1% |
| GRAPE | 2,136×2,136 | .756 | 62.6 | 20.0 | 31.4 | 1.4% |
| IDRiD | 4,288×2,848 | .692 | 87.0 | 17.8 | 47.1 | 1.2% |
| Jichi | 848×848 | .788 | 99.4 | 17.1 | 84.2 | 1.2% |
| SUSTech-SYSU | 2,880×2,136 | .896 | 109.9 | 13.7 | 54.7 | 1.1% |
| AFIO | 1,612×1,536 | .651 | 55.7 | 15.8 | 70.8 | 3.1% |
| FIVES | 2,048×2,048 | .758 | 53.2 | 17.0 | 52.2 | 3.2% |
| HRF | 3,504×2,336 | .845 | 92.7 | 17.4 | 44.1 | 2.2% |
| HYGD | 1,897×1,897 | .787 | 62.2 | 20.5 | 11.0 | 1.5% |
| PALM | 2,124×2,056 | .804 | 55.4 | 17.0 | 28.3 | 0.0% |
| PAPILA | 2,576×1,934 | .809 | 61.7 | 13.3 | 26.1 | 1.9% |
| Paraguay | 2,124×2,056 | .805 | 39.0 | 10.2 | 16.2 | 1.0% |
| Chákṣu | 2,448×3,264 | .465 | 55.6 | 24.6 | 211.3 | 1.0% |

*Reading.* Sources differ strongly in resolution, field of view, exposure and sharpness, and these trivial statistics alone identify the source; flagged images are rare (0–3%). The hardest test sources are not simply the lowest-quality ones: AFIO and FIVES are dark, but Paraguay, the darkest and among the least sharp, is the easiest for every method. Source difficulty therefore reflects class composition and label scope as well as appearance, and we do not attribute it to image quality alone.

### S1.4 Mapping from source labels

Table S4: Native labels mapped to each harmonized class (first token of the stored label; device and vote suffixes omitted).

| Source | Class | Native labels |
|---|---|---|
| Bangladesh Eye | N | Healthy |
| Bangladesh Eye | D | Diabetic_Retinopathy |

Table S4: (continued)

| Source | Class | Native labels |
|---|---|---|
| Bangladesh Eye | G | Glaucoma |
| OIA-DDR | N | ddr_0 |
| OIA-DDR | D | ddr_1, ddr_2, ddr_3, ddr_4 |
| DeepDRiD | N | icdr_0 |
| DeepDRiD | D | icdr_1, icdr_2, icdr_3, icdr_4 |
| GRAPE | G | glaucoma_follow_up |
| IDRiD | N | icdr_0 |
| IDRiD | D | icdr_1, icdr_2, icdr_3, icdr_4 |
| Jichi | N | davis_ndr |
| Jichi | D | davis_pdr, davis_ppdr, davis_sdr |
| SUSTech-SYSU | N | icdr_0 |
| SUSTech-SYSU | D | icdr_1, icdr_2, icdr_3, icdr_4, laser_scars(reclassified_icdr_2), laser_scars(reclassified_icdr_3), ... |
| AFIO | N | macula_healthy, od_healthy |
| AFIO | D | macula_ci-dme |
| AFIO | G | od_glaucoma |
| FIVES | N | normal |
| FIVES | D | diabetic_retinopathy |
| FIVES | G | glaucoma |
| HRF | N | healthy |
| HRF | D | diabetic_retinopathy |
| HRF | G | glaucoma |
| HYGD | G | GON+ |
| PALM | N | normal |
| PAPILA | N | healthy |
| PAPILA | G | glaucoma |
| Paraguay | N | 1._No_DR_signs |
| Paraguay | D | 2._Mild_(or_early)_NPDR, 3._Moderate_NPDR, 4._Severe_NPDR, 5._Very_Severe_NPDR, 6._PDR, 7._Advanced_PDR |
| Chákṣu | N | normal |
| Chákṣu | G | glaucoma_suspect |

*Reading.* A native label is mapped only when it supports the class. Grading scales differ: ICDR grades 1–4, Davis SDR–PDR and named DR categories all become D. Glaucoma suspects in Chákṣu follow the source's majority vote. HYGD's 199 GON− images were excluded rather than mapped to N, because "no glaucomatous optic neuropathy" does not certify the absence of other disease; this makes HYGD single-class. The glaucoma reference standard differs between sources: clinical work-up in GRAPE (visual-field follow-up patients) and HYGD (GON+), image-based expert grading in Chákṣu (majority vote, including suspects), and not reported for Bangladesh Eye and AFIO, so G pools definitions of different strength. D pools all DR grades from mild NPDR upward; referable DR is not separated (recall by grade: Table S42). The 861-image open-set probe holds images of other conditions from PALM (637), FIVES (200) and AFIO (24), and is distinct from the 860 RFMiD 2.0 images excluded from the core task.

### S1.5 Leakage control and audit

Duplicates were detected by exact SHA-256 hashes and by a vessel-pattern signature (band-pass green channel, $32 \times 32$) with shift-tolerant normalized cross-correlation at threshold 0.70, then joined with patient identifiers into leakage groups. An automated audit of the released splits found zero violations for each of: a leakage group in more than one split; an identical file in more than one split; a patient in more than one split;

a near-duplicate pair across splits; an acquisition family on both development and test sides; a source on both sides; a calibration image without a Cal-A/Cal-B assignment. Brightness-altered copies of GRAPE photographs inside the FIVES glaucoma class were found this way and removed. The detector linked 4,247 image pairs (678 byte-identical, 29 across sources), forming 2,450 leakage groups with more than one image. Of 40,414 inventoried images, 35,280 enter the protocol; the rest were removed for an excluded or unmapped native label (1,820 and 1,453), duplicates whose copies carried conflicting labels (956), a source excluded from the core task (860), or duplication of a test-side image (45). The code is `fsbench/dedup.py` and `fsbench/vessel.py`.

**Table S5.** Duplicate links found by the detector (exact hash or vessel-pattern NCC $\geq 0.70$), by source pair: links, byte-identical links, and links whose two images carry different labels.

| Scope | Source(s) | Links | Exact | Label conflict |
|---|---|---|---|---|
| across | FIVES – GRAPE | 29 | 0 | 0 |
| within | Jichi | 2,691 | 0 | 316 |
| within | Bangladesh Eye | 770 | 484 | 612 |
| within | OIA-DDR | 203 | 106 | 1 |
| within | GRAPE | 189 | 0 | 0 |
| within | Paraguay | 181 | 19 | 3 |
| within | SUSTech-SYSU | 55 | 0 | 0 |
| within | DeepDRiD | 38 | 0 | 0 |
| within | PALM | 22 | 21 | 6 |
| within | FIVES | 18 | 6 | 7 |
| within | AFIO | 16 | 16 | 12 |
| within | HYGD | 11 | 10 | 2 |
| within | RFMiD 2.0 | 10 | 3 | 4 |
| within | Chákṣu | 8 | 7 | 1 |
| within | IDRiD | 6 | 6 | 1 |

*Reading.* All 29 cross-source links join GRAPE photographs to brightness-altered copies inside FIVES, a development–test leak that hashes alone would miss. Within-source links mostly join repeated visits or re-exports; links with conflicting labels concentrate in Bangladesh Eye and Jichi, and the 956 images removed for conflicting labels (above) come from such groups.

**Excluded collections.** RFMiD 2.0 was used for inventory and duplicate checks only, because class is confounded with camera. A glaucoma collection of optic-disc crops was excluded for its different field of view. OIA-DDR is CC BY-NC-SA 4.0, so derived data remain non-commercial.

**Ethics.** All images are publicly released; this is a secondary analysis. Ethics and consent statements were found in the publications of FIVES, GRAPE, PAPILA, SUSTech-SYSU, DeepDRiD, Jichi, HYGD, Chákṣu, PALM and Paraguay. Statements were not found on the release pages of HRF, Bangladesh Eye and AFIO. Age and sex are available only for PAPILA, so subgroup fairness cannot be assessed.

## S2 Models, Recipes and Prompts

**Table S6.** Encoders. "Fine-tuned" marks the 13 architectures trained end to end; every encoder was also probed frozen.

| Encoder | Pretraining data | Fine-tuned |
|---|---|---|
| deit_b16 | ImageNet-1k | yes |
| convnext_t | ImageNet-1k | yes |
| swin_t | ImageNet-1k | yes |
| inception_v3 | ImageNet-1k | yes |
| efficientnet_b0 | ImageNet-1k | yes |
| densenet121 | ImageNet-1k | yes |
| vit_b16_in21k | ImageNet-21k | yes |
| resnet50 | ImageNet-1k | yes |
| mobilenetv3_l | ImageNet-1k | yes |
| vgg16 | ImageNet-1k | yes |
| vit_b16_mae | ImageNet-1k (MAE, self-supervised) | yes |
| dinov2_b14 | LVD-142M (self-supervised) | no |
| retfound_mae | MEH-MIDAS + public fundus (MAE, RETFound) | yes |
| retfound_dinov2 | MEH-MIDAS fundus (DINOv2, RETFound) | yes |
| clip_b16 | OpenAI web image-text | no |
| metaclip_b16 | MetaCLIP (web image-text) | no |
| siglip_b16 | WebLI | no |
| siglip_l16 | WebLI | no |
| siglip2_b32 | WebLI | no |
| siglip2_b16 | WebLI | no |
| siglip2_l16 | WebLI | no |
| bioclip_b16 | iNat/biodiversity (TreeOfLife-10M) | no |
| bioclip2_l14 | biodiversity (TreeOfLife-200M) | no |
| biomedclip | PMC-15M biomedical figures | no |
| medsiglip | Medical image-text (MedGemma report); fundus: EyePACS | no |
| flair | Retinal image-text; includes five FundusShift test sources | no |

**Table S7.** Training recipes. Every setting selects by validation bAcc unless stated; post-hoc calibration always uses Cal-A, thresholds Cal-B.

| | |
|---|---|
| Fine-tuning | Native input size and normalization; AdamW, lr $10^{-4}$ (head $10^{-3}$), wd 0.05, cosine with one warm-up epoch, 15 epochs, batch 32; (source, class)-balanced sampling; crop, flip, rotation and colour augmentation; mixed precision; five losses (CE; label smoothing 0.1; focal $\gamma$=3; MDCA $\beta$=1; MbLS $m$=10, $\lambda$=0.1); 3 seeds. NLL-rule arm: checkpoint by validation NLL. |
| Linear, MLP probes | Frozen embeddings standardized on train; AdamW lr $10^{-3}$, wd $10^{-4}$, 40 epochs, batch 256; weighting none, class or (class, source); MLP: one hidden layer of 512. |
| 16-shot probe | 16 images per class, one per leakage group, redrawn per seed, 300 epochs. |
| Zero-shot | Three prompt banks (Table S8); bank chosen by validation bAcc. |
| CoOp | 16 learnable context tokens, frozen towers, SGD lr 0.002, cosine, 200 full-batch steps on the 16 shots; ctx-init: 4 tokens from "a photo of a"; CoOp+LP: then a linear head, 200 steps, lr $10^{-3}$; loss variants CE, focal, LS, MbLS, MDCA. |
| Tip-Adapter | Official recipe on the same shots; $(\alpha, \beta)$ grid and bank by validation bAcc (official accuracy rule kept as sensitivity); -F: keys fine-tuned 20 epochs, AdamW lr $10^{-3}$. |
| ERM, Group DRO, CORAL | Fine-tuning recipe with ConvNeXt-T, or DINOv2-B/14 with LoRA rank 8 on query and value; Group DRO $\eta$=0.01, group = development source; CORAL $\lambda$=1. |
| SWAD | Dense averaging over segments of 150 updates, official loss-valley rule (3, 6, 0.3); replaces the validation checkpoint. |
| WiSE-FT | Head from the zero-shot classifier, AdamW lr $10^{-5}$, wd 0.1, 15 epochs; interpolation $\alpha \in \{0, 0.1, \ldots, 1\}$ chosen by validation bAcc. |
| Multiple choice | Image plus a three-option question; next-token log-likelihoods of the option letters averaged over all six option orders. |
| LOTUS | Table S44. |

Table S8: Prompt banks used by every zero-shot model and by the text branch of LOTUS.

| Bank | Prompts |
|---|---|
| CLIP ensemble | a photo of .; a colour fundus photograph of .; a retinal image of .; a fundus photo showing .; an ophthalmology image of .; a close-up retinal photograph of .; . with class names: N: a normal retina, D: diabetic retinopathy, G: glaucoma |
| CuPL (N) | a healthy fundus with a pink optic disc, a clear macula and regular retinal vessels. a normal retina without haemorrhages, exudates or optic disc abnormality. a fundus photograph with an evenly coloured background and a well-defined foveal reflex. a retina with a small physiological optic cup and an intact neural rim. a fundus free of microaneurysms, cotton-wool spots and neovascularisation. |
| CuPL (D) | a retina with scattered microaneurysms and dot-blot haemorrhages. a fundus with yellow hard exudates near the macula. a retinal photograph with cotton-wool spots and venous beading. a fundus with abnormal new vessels and preretinal haemorrhage. a diabetic eye with intraretinal haemorrhages in several quadrants. |
| CuPL (G) | an optic disc with a large cup-to-disc ratio. a fundus with a thinned or notched neural rim of the optic disc. a retina with retinal nerve fibre layer defects next to the optic disc. an optic nerve head with vertical cupping and pallor. a fundus with peripapillary atrophy and glaucomatous optic neuropathy. |
| Single (N) | a colour fundus photograph of a healthy retina with a normal optic disc and no lesions. |
| Single (D) | a colour fundus photograph with diabetic retinopathy showing microaneurysms, haemorrhages and exudates. |
| Single (G) | a colour fundus photograph with glaucoma showing an enlarged optic cup and a thinned neural rim. |

# S3 Complete Benchmark Results

Table S9: Every setting and encoder, with the configuration of highest validation bAcc. †FLAIR's pretraining includes five FundusShift test sources. Multiple-choice models have no validation scores and no probabilities.

| Setting | Encoder | Config | Val | Test | Eq-src | Chákṣu | ECE R/T/F | NLL R/T/F |
|---|---|---|---|---|---|---|---|---|
| 16-shot probe | flair† | none/mbls | .809 | .750 | .684 | .683 | .134/.128/.127 | .947/.915/.891 |
| 16-shot probe | medsiglip | none/ce | .802 | .748 | .683 | .675 | .096/.044/.037 | .701/.644/.644 |
| 16-shot probe | siglip2_l16 | none/mbls | .696 | .621 | .564 | .195 | .124/.075/.074 | .999/.885/.889 |
| 16-shot probe | biomedclip | none/ce | .677 | .479 | .430 | .450 | .198/.113/.120 | 1.18/1.05/1.06 |
| 16-shot probe | dinov2_b14 | none/ce | .675 | .529 | .503 | .192 | .275/.080/.088 | 1.53/.993/.996 |
| 16-shot probe | efficientnet_b0 | none/ce | .665 | .482 | .458 | .279 | .216/.081/.085 | 1.24/1.02/1.03 |
| 16-shot probe | retfound_mae | none/mbls | .662 | .373 | .339 | .207 | .437/.238/.241 | 1.94/1.17/1.17 |
| 16-shot probe | siglip_l16 | none/ce | .658 | .440 | .414 | .316 | .314/.166/.168 | 1.51/1.16/1.16 |
| 16-shot probe | siglip_b16 | none/ce | .653 | .526 | .516 | .266 | .175/.093/.102 | 1.24/1.02/1.03 |
| 16-shot probe | densenet121 | none/ce | .653 | .514 | .499 | .347 | .207/.063/.056 | 1.15/.963/.955 |
| 16-shot probe | convnext_t | none/ce | .649 | .523 | .520 | .403 | .214/.063/.069 | 1.24/.971/.970 |
| 16-shot probe | clip_b16 | none/mbls | .649 | .486 | .419 | .251 | .192/.103/.108 | 1.24/1.08/1.09 |
| 16-shot probe | deit_b16 | none/mbls | .645 | .517 | .484 | .233 | .228/.096/.097 | 1.24/1.01/1.01 |
| 16-shot probe | retfound_dinov2 | none/ce | .644 | .437 | .426 | .158 | .282/.087/.095 | 1.47/1.09/1.09 |
| 16-shot probe | mobilenetv3_l | none/ce | .643 | .450 | .426 | .357 | .277/.106/.113 | 1.48/1.08/1.09 |
| 16-shot probe | siglip2_b16 | none/ce | .638 | .456 | .422 | .281 | .315/.137/.141 | 1.85/1.11/1.11 |
| 16-shot probe | swin_t | none/ce | .634 | .494 | .458 | .204 | .214/.059/.062 | 1.21/1.00/1.01 |
| 16-shot probe | vit_b16_in21k | none/ce | .630 | .495 | .469 | .268 | .234/.087/.090 | 1.34/1.02/1.02 |
| 16-shot probe | resnet50 | none/mbls | .628 | .541 | .509 | .429 | .149/.070/.061 | 1.12/.986/.979 |
| 16-shot probe | inception_v3 | none/mbls | .627 | .457 | .425 | .437 | .323/.099/.101 | 1.59/1.07/1.08 |
| 16-shot probe | siglip2_b32 | none/ce | .622 | .471 | .439 | .375 | .260/.098/.098 | 1.41/1.07/1.07 |
| 16-shot probe | vit_b16_mae | none/ls | .621 | .392 | .412 | .239 | .331/.189/.200 | 1.57/1.20/1.20 |
| 16-shot probe | vgg16 | none/ls | .619 | .468 | .459 | .378 | .213/.135/.125 | 1.73/1.24/1.19 |
| 16-shot probe | metaclip_b16 | none/ce | .613 | .612 | .539 | .458 | .145/.085/.079 | 1.10/.906/.903 |
| 16-shot probe | bioclip2_l14 | none/ce | .612 | .495 | .467 | .407 | .247/.064/.068 | 1.48/1.02/1.03 |
| 16-shot probe | bioclip_b16 | none/ce | .597 | .411 | .386 | .204 | .347/.142/.141 | 1.64/1.14/1.14 |
| CoOp | siglip_b16 | ce | .660 | .491 | .458 | .318 | .236/.175/.177 | 1.35/1.18/1.16 |
| CoOp | biomedclip | ls | .654 | .475 | .437 | .379 | .235/.163/.173 | 1.37/1.19/1.21 |
| CoOp | clip_b16 | ls | .631 | .448 | .382 | .206 | .235/.189/.190 | 1.27/1.16/1.17 |
| CoOp | bioclip2_l14 | mbls | .622 | .474 | .462 | .270 | .235/.133/.134 | 1.50/1.21/1.24 |
| CoOp | siglip_l16 | ce | .612 | .403 | .479 | .368 | .107/.189/.191 | 1.13/1.25/1.28 |
| CoOp | bioclip_b16 | mdca | .593 | .415 | .422 | .152 | .429/.154/.155 | 2.59/1.19/1.20 |
| CoOp | metaclip_b16 | ls | .585 | .540 | .483 | .358 | .125/.089/.084 | .990/.951/.944 |
| CoOp | siglip2_l16 | zs_penalty | .408 | .459 | .489 | .444 | .099/.044/.045 | 1.01/.994/.997 |
| CoOp | siglip2_b32 | zs_penalty | .401 | .399 | .315 | .427 | .045/.191/.186 | 1.05/1.11/1.10 |
| CoOp | siglip2_b16 | zs_penalty | .358 | .335 | .233 | .205 | .132/.224/.224 | 1.10/1.16/1.16 |
| CoOp ctx-init | biomedclip | mdca | .642 | .497 | .447 | .384 | .287/.150/.160 | 1.76/1.19/1.22 |
| CoOp ctx-init | clip_b16 | mdca | .638 | .499 | .429 | .171 | .189/.133/.134 | 1.20/1.08/1.08 |
| CoOp ctx-init | metaclip_b16 | mdca | .634 | .605 | .534 | .418 | .092/.043/.042 | .884/.838/.836 |
| CoOp ctx-init | bioclip2_l14 | mdca | .611 | .468 | .458 | .231 | .188/.115/.115 | 1.36/1.18/1.20 |
| CoOp ctx-init | bioclip_b16 | mbls | .583 | .444 | .418 | .157 | .419/.173/.175 | 2.56/1.21/1.22 |
| CoOp ctx-init | siglip_l16 | zs_penalty | .442 | .421 | .519 | .480 | .075/.092/.097 | 1.08/1.09/1.09 |
| CoOp ctx-init | siglip_b16 | mdca | .421 | .458 | .513 | .492 | .048/.120/.119 | 1.09/1.06/1.06 |
| CoOp ctx-init | siglip2_l16 | mdca | .386 | .361 | .450 | .499 | .231/.183/.178 | 1.29/1.20/1.23 |
| CoOp ctx-init | siglip2_b32 | zs_penalty | .358 | .380 | .447 | .518 | .054/.061/.058 | 1.06/1.06/1.06 |

Table S9: (continued)

| Setting | Encoder | Config | Val | Test | Eq-src | Chákṣu | ECE R/T/F | NLL R/T/F |
|---|---|---|---|---|---|---|---|---|
| CoOp ctx-init | siglip2_b16 | focal | .351 | .341 | .424 | .496 | .178/.157/.159 | 1.24/1.20/1.22 |
| CoOp + LP | siglip_b16 | mbls | .668 | .496 | .463 | .314 | .243/.176/.179 | 1.37/1.18/1.17 |
| CoOp + LP | biomedclip | mdca | .652 | .482 | .433 | .368 | .292/.177/.186 | 1.68/1.22/1.24 |
| CoOp + LP | siglip_l16 | ce | .647 | .447 | .490 | .327 | .101/.169/.172 | 1.12/1.23/1.27 |
| CoOp + LP | clip_b16 | ls | .637 | .449 | .384 | .213 | .244/.190/.191 | 1.29/1.16/1.16 |
| CoOp + LP | siglip2_l16 | ls | .632 | .559 | .543 | .276 | .154/.030/.030 | .969/.897/.895 |
| CoOp + LP | metaclip_b16 | mbls | .628 | .594 | .506 | .410 | .170/.067/.066 | .959/.860/.857 |
| CoOp + LP | bioclip2_l14 | mbls | .623 | .477 | .463 | .276 | .253/.136/.136 | 1.59/1.21/1.23 |
| CoOp + LP | bioclip_b16 | mdca | .595 | .415 | .424 | .156 | .433/.156/.157 | 2.63/1.19/1.20 |
| CoOp + LP | siglip2_b32 | ce | .539 | .487 | .432 | .459 | .068/.064/.061 | 1.00/.998/.995 |
| CoOp + LP | siglip2_b16 | mbls | .510 | .419 | .369 | .417 | .051/.075/.073 | 1.04/1.04/1.04 |
| CORAL | convnext_t | ce | .883 | .676 | .623 | .349 | .198/.136/.148 | 1.38/1.01/1.11 |
| CORAL | dinov2_lora | ce | .853 | .667 | .625 | .340 | .143/.144/.150 | 1.05/1.05/1.11 |
| ERM | dinov2_lora | ce | .857 | .707 | .646 | .419 | .154/.115/.119 | 1.12/.944/.965 |
| Fine-tune | swin_t | mbls | .893 | .655 | .600 | .298 | .223/.171/.182 | 1.56/1.17/1.31 |
| Fine-tune | convnext_t | ls | .888 | .676 | .611 | .319 | .144/.168/.177 | .926/.982/1.04 |
| Fine-tune | retfound_mae | mdca | .887 | .525 | .417 | .291 | .370/.276/.288 | 2.69/1.50/1.57 |
| Fine-tune | retfound_dinov2 | mbls | .886 | .583 | .501 | .358 | .285/.192/.207 | 1.93/1.19/1.33 |
| Fine-tune | densenet121 | ce | .882 | .626 | .581 | .309 | .238/.139/.158 | 1.88/1.13/1.30 |
| Fine-tune | resnet50 | mdca | .881 | .666 | .620 | .319 | .184/.110/.126 | 1.48/1.05/1.17 |
| Fine-tune | inception_v3 | mbls | .877 | .664 | .624 | .405 | .200/.139/.147 | 1.25/.956/1.05 |
| Fine-tune | vgg16 | mbls | .875 | .671 | .625 | .371 | .169/.135/.147 | 1.28/1.10/1.24 |
| Fine-tune | deit_b16 | mdca | .868 | .622 | .560 | .313 | .230/.158/.167 | 1.88/1.28/1.42 |
| Fine-tune | vit_b16_in21k | mdca | .863 | .596 | .535 | .336 | .251/.180/.190 | 2.02/1.31/1.42 |
| Fine-tune | efficientnet_b0 | ce | .861 | .651 | .596 | .266 | .188/.108/.122 | 1.34/.974/1.04 |
| Fine-tune | vit_b16_mae | ce | .856 | .587 | .532 | .302 | .272/.196/.205 | 2.17/1.42/1.59 |
| Fine-tune | mobilenetv3_l | ls | .846 | .648 | .596 | .238 | .082/.130/.144 | .855/.923/.993 |
| Fine-tune, NLL rule | convnext_t | ce | .870 | .677 | .619 | .324 | .185/.133/.147 | 1.35/1.04/1.16 |
| Fine-tune, NLL rule | resnet50 | ce | .857 | .647 | .601 | .254 | .192/.117/.133 | 1.57/1.12/1.28 |
| Group DRO | convnext_t | ce | .849 | .707 | .649 | .417 | .082/.091/.101 | .886/.926/1.04 |
| Group DRO | dinov2_lora | ce | .798 | .733 | .677 | .486 | .050/.035/.034 | .732/.707/.732 |
| Linear probe | medsiglip | class/ls | .920 | .728 | .654 | .347 | .024/.079/.093 | .647/.694/.739 |
| Linear probe | flair† | class/focal | .900 | .721 | .693 | .703 | .088/.117/.117 | .705/.777/.775 |
| Linear probe | dinov2_b14 | class/mbls | .889 | .674 | .616 | .434 | .104/.103/.115 | .843/.842/.875 |
| Linear probe | retfound_mae | class/ls | .886 | .455 | .351 | .243 | .368/.412/.414 | 1.43/1.71/1.69 |
| Linear probe | retfound_dinov2 | class/ce | .868 | .606 | .522 | .115 | .185/.147/.158 | 1.16/1.03/1.08 |
| Linear probe | siglip2_l16 | class/mbls | .866 | .605 | .562 | .206 | .167/.159/.170 | 1.11/1.08/1.16 |
| Linear probe | vit_b16_mae | class/ls | .866 | .579 | .536 | .102 | .136/.190/.200 | .924/1.02/1.03 |
| Linear probe | vit_b16_in21k | class/ls | .861 | .608 | .572 | .292 | .119/.170/.182 | .983/1.09/1.15 |
| Linear probe | metaclip_b16 | class/ls | .861 | .599 | .551 | .101 | .114/.177/.188 | .908/1.01/1.05 |
| Linear probe | siglip2_b16 | class/ls | .860 | .546 | .520 | .484 | .130/.187/.201 | 1.04/1.15/1.22 |
| Linear probe | siglip_b16 | class/ls | .856 | .567 | .540 | .169 | .130/.189/.201 | .988/1.10/1.13 |
| Linear probe | bioclip2_l14 | class/ls | .853 | .538 | .520 | .365 | .164/.212/.227 | 1.05/1.15/1.17 |
| Linear probe | densenet121 | class/mdca | .852 | .569 | .522 | .121 | .228/.203/.211 | 1.38/1.26/1.35 |
| Linear probe | siglip_l16 | class/ce | .851 | .524 | .508 | .352 | .237/.235/.246 | 1.46/1.45/1.57 |
| Linear probe | swin_t | class/mbls | .850 | .584 | .523 | .176 | .198/.180/.191 | 1.20/1.14/1.17 |
| Linear probe | convnext_t | class/mbls | .848 | .615 | .553 | .218 | .145/.137/.147 | .999/.981/1.01 |
| Linear probe | clip_b16 | class/ls | .844 | .595 | .528 | .477 | .104/.162/.171 | .950/1.07/1.14 |
| Linear probe | deit_b16 | class/mdca | .844 | .595 | .537 | .272 | .179/.155/.167 | 1.11/1.05/1.09 |
| Linear probe | bioclip_b16 | class/mbls | .830 | .539 | .479 | .164 | .207/.198/.212 | 1.22/1.19/1.23 |
| Linear probe | mobilenetv3_l | class/ls | .825 | .583 | .525 | .137 | .089/.153/.165 | .968/1.07/1.11 |
| Linear probe | siglip2_b32 | class/ls | .824 | .554 | .517 | .461 | .130/.177/.188 | 1.01/1.10/1.13 |
| Linear probe | efficientnet_b0 | class/ce | .824 | .598 | .534 | .196 | .210/.179/.185 | 1.32/1.19/1.18 |
| Linear probe | resnet50 | class/mbls | .815 | .650 | .576 | .334 | .081/.056/.060 | .888/.858/.868 |
| Linear probe | biomedclip | class/mdca | .803 | .479 | .459 | .510 | .274/.271/.281 | 1.34/1.33/1.36 |
| Linear probe | inception_v3 | class/mbls | .799 | .543 | .500 | .347 | .220/.190/.198 | 1.22/1.14/1.15 |
| Linear probe | vgg16 | class/mbls | .793 | .534 | .480 | .267 | .234/.157/.179 | 1.89/1.56/1.36 |
| MLP probe | dinov2_b14 | none/ce | .870 | .598 | .524 | .318 | .263/.172/.183 | 2.31/1.26/1.42 |
| MLP probe | siglip2_l16 | none/ce | .859 | .559 | .481 | .140 | .281/.194/.194 | 2.92/1.73/1.73 |
| MLP probe | metaclip_b16 | none/ce | .853 | .524 | .473 | .122 | .360/.260/.269 | 3.04/1.53/1.69 |
| MLP probe | siglip_b16 | none/ce | .851 | .570 | .542 | .330 | .279/.182/.189 | 2.71/1.42/1.57 |
| MLP probe | siglip_l16 | none/ce | .847 | .546 | .469 | .355 | .346/.212/.216 | 4.72/1.86/2.02 |
| MLP probe | siglip2_b16 | none/ce | .845 | .542 | .510 | .431 | .292/.205/.217 | 2.95/1.71/2.05 |
| MLP probe | bioclip2_l14 | none/ce | .838 | .514 | .457 | .318 | .349/.234/.245 | 3.01/1.53/1.66 |
| MLP probe | clip_b16 | none/ce | .837 | .545 | .487 | .450 | .344/.250/.256 | 4.01/1.94/2.18 |
| MLP probe | bioclip_b16 | none/ce | .827 | .502 | .434 | .089 | .328/.252/.255 | 2.14/1.50/1.56 |
| MLP probe | siglip2_b32 | none/ce | .817 | .546 | .503 | .490 | .302/.181/.189 | 3.26/1.52/1.74 |
| MLP probe | biomedclip | none/ce | .806 | .525 | .506 | .475 | .272/.185/.194 | 2.84/1.85/2.08 |
| SWAD | convnext_t | ce | .883 | .672 | .622 | .309 | .208/.139/.149 | 1.57/1.06/1.16 |
| SWAD | dinov2_lora | ce | .832 | .716 | .655 | .436 | .127/.092/.097 | .964/.854/.879 |
| Tip-Adapter | biomedclip | – | .597 | .478 | .491 | .617 | .332/.091/.096 | 1.75/1.02/1.03 |
| Tip-Adapter | siglip2_l16 | – | .550 | .497 | .534 | .533 | .140/.128/.117 | 1.00/.930/.914 |
| Tip-Adapter | clip_b16 | – | .473 | .456 | .466 | .424 | .082/.071/.077 | 1.04/1.08/1.07 |
| Tip-Adapter (acc. rule) | biomedclip | – | .597 | .478 | .491 | .617 | .332/.091/.096 | 1.75/1.02/1.03 |
| Tip-Adapter (acc. rule) | siglip2_l16 | – | .550 | .497 | .534 | .533 | .140/.128/.117 | 1.00/.930/.914 |
| Tip-Adapter (acc. rule) | clip_b16 | – | .473 | .456 | .466 | .424 | .082/.071/.077 | 1.04/1.08/1.07 |
| Tip-Adapter-F | biomedclip | – | .666 | .495 | .453 | .532 | .225/.147/.152 | 1.28/1.10/1.11 |
| Tip-Adapter-F | siglip2_l16 | – | .602 | .538 | .551 | .528 | .086/.157/.155 | 1.09/.986/.969 |
| Tip-Adapter-F | clip_b16 | – | .564 | .508 | .511 | .459 | .109/.095/.090 | 1.05/.997/.995 |
| Tip-Adapter-F (acc. rule) | biomedclip | – | .666 | .495 | .453 | .532 | .225/.147/.152 | 1.28/1.10/1.11 |
| Tip-Adapter-F (acc. rule) | siglip2_l16 | – | .602 | .538 | .551 | .528 | .086/.157/.155 | 1.09/.986/.969 |
| Tip-Adapter-F (acc. rule) | clip_b16 | – | .564 | .508 | .511 | .459 | .109/.095/.090 | 1.05/.997/.995 |
| WiSE-FT | siglip2_l16 | fully-tuned | .886 | .634 | .587 | .406 | .236/.168/.179 | 1.84/1.24/1.38 |
| Zero-shot | flair† | descriptive_single | .653 | .712 | .624 | .567 | .162/.154/.137 | 1.09/.867/.827 |
| Zero-shot | medsiglip | descriptive_single | .650 | .709 | .606 | .615 | .207/.032/.032 | .838/.690/.690 |
| Zero-shot | biomedclip | descriptive_single | .503 | .471 | .535 | .562 | .229/.061/.074 | 1.29/1.00/1.02 |
| Zero-shot | metaclip_b16 | descriptive_single | .476 | .403 | .406 | .374 | .093/.119/.119 | 1.06/1.07/1.07 |

Table S9: (continued)

| Setting | Encoder | Config | Val | Test | Eq-src | Chákṣu | ECE R/T/F | NLL R/T/F |
|---|---|---|---|---|---|---|---|---|
| Zero-shot | siglip2_b32 | descriptive_single | .473 | .463 | .501 | .505 | .028/.069/.063 | 1.05/1.07/1.07 |
| Zero-shot | siglip2_l16 | clip_ensemble | .461 | .481 | .494 | .374 | .031/.070/.085 | 1.06/1.17/1.13 |
| Zero-shot | siglip_b16 | cupl | .437 | .433 | .508 | .503 | .068/.090/.092 | .959/.975/.973 |
| Zero-shot | siglip2_b16 | descriptive_single | .418 | .509 | .539 | .518 | .107/.176/.167 | 1.06/1.07/1.06 |
| Zero-shot | bioclip_b16 | descriptive_single | .366 | .457 | .518 | .525 | .358/.182/.183 | 3.79/1.10/1.10 |
| Zero-shot | siglip_l16 | cupl | .364 | .357 | .453 | .500 | .280/.162/.165 | 1.24/1.09/1.10 |
| Zero-shot | bioclip2_l14 | clip_ensemble | .344 | .362 | .281 | .074 | .326/.180/.186 | 1.54/1.16/1.18 |
| Zero-shot | clip_b16 | clip_ensemble | .327 | .394 | .445 | .173 | .055/.096/.094 | 1.10/1.15/1.16 |
| Multiple choice | Fundus-R1-7B | avg6 | – | .576 | .569 | .463 | – | – |
| Multiple choice | Qwen2.5-VL-7B-Instruct | avg6 | – | .424 | .462 | .488 | – | – |
| Multiple choice | llava-med-v1.5-mistral-7b-hf | avg6 | – | .347 | .448 | .507 | – | – |
| Multiple choice | llava-onevision-qwen2-7b-ov-hf | avg6 | – | .407 | .327 | .096 | – | – |

*Reading.* Within each setting the encoder ranking on test differs from that on validation (Sect. S5). MedSigLIP is the strongest frozen encoder across probe settings, and FLAIR is comparable only on sources it did not see in pretraining. Fine-tuned RETFound reaches high validation (0.886–0.887) but 0.525–0.583 on test, despite fundus pretraining. TS reduces ECE and NLL for most fine-tuned models but leaves ECE far above that of frozen probes. For transparency, the best frozen probe judged by test itself, a class-weighted cross-entropy MedSigLIP linear probe, reached test bAcc 0.778, equal-source 0.702, raw ECE 0.065 and recall G 0.657; validation selected the label-smoothed variant instead.

## S4 Per-Source and Per-Class Results

Table S10: bAcc per unseen source (raw), equal-source mean and worst test source, for LOTUS and the validation-selected representatives.

| Method | Paraguay | HYGD | FIVES | PAPILA | PALM | HRF | AFIO | Chákṣu | Eq-src | Worst |
|---|---|---|---|---|---|---|---|---|---|---|
| LOTUS | .919 | .822 | .721 | .664 | .836 | .622 | .652 | .695 | .748 | .622 |
| 16-shot probe, medsiglip | .874 | .756 | .668 | .499 | .723 | .711 | .548 | .675 | .683 | .499 |
| CoOp, siglip_b16 | .728 | .130 | .489 | .197 | .768 | .415 | .476 | .318 | .458 | .130 |
| CoOp ctx-init, biomedclip | .753 | .038 | .481 | .435 | .563 | .400 | .456 | .384 | .447 | .038 |
| CoOp + LP, siglip_b16 | .735 | .131 | .496 | .197 | .781 | .430 | .471 | .314 | .463 | .131 |
| CORAL, convnext_t | .929 | .263 | .688 | .413 | .911 | .622 | .533 | .349 | .623 | .263 |
| ERM, dinov2_lora | .884 | .505 | .596 | .498 | .816 | .607 | .614 | .419 | .646 | .498 |
| Fine-tune, swin_t | .913 | .237 | .647 | .468 | .836 | .600 | .501 | .298 | .600 | .237 |
| Fine-tune, NLL rule, convnext_t | .917 | .283 | .704 | .449 | .880 | .585 | .516 | .324 | .619 | .283 |
| Group DRO, convnext_t | .920 | .316 | .704 | .592 | .870 | .600 | .539 | .417 | .649 | .316 |
| Linear probe, medsiglip | .934 | .477 | .559 | .493 | .907 | .637 | .571 | .347 | .654 | .477 |
| MLP probe, dinov2_b14 | .889 | .092 | .631 | .285 | .747 | .533 | .492 | .318 | .524 | .092 |
| SWAD, convnext_t | .928 | .243 | .709 | .409 | .901 | .615 | .550 | .309 | .622 | .243 |
| Tip-Adapter, biomedclip | .399 | .744 | .427 | .496 | .480 | .400 | .493 | .617 | .491 | .399 |
| Tip-Adapter-F, biomedclip | .624 | .283 | .465 | .471 | .472 | .407 | .448 | .532 | .453 | .283 |
| WiSE-FT, siglip2_l16 | .881 | .250 | .626 | .423 | .801 | .600 | .529 | .406 | .587 | .250 |
| Zero-shot, medsiglip | .729 | .719 | .723 | .622 | .379 | .556 | .512 | .615 | .606 | .379 |

Table S11: Class recalls on the multi-class test sources.

| | FIVES | | | AFIO | | | PAPILA | | Paraguay | |
|---|---|---|---|---|---|---|---|---|---|---|
| Method | N | D | G | N | D | G | N | G | N | D |
| LOTUS | .606 | .767 | .789 | .448 | .833 | .674 | .899 | .429 | .893 | .946 |

Table S11: (continued)

| Method | FIVES | | | AFIO | | | PAPILA | | Paraguay | |
|---|---|---|---|---|---|---|---|---|---|---|
| | N | D | G | N | D | G | N | G | N | D |
| 16-shot probe, medsiglip | .418 | .757 | .831 | .075 | .796 | .773 | .139 | .858 | .797 | .951 |
| CoOp, siglip_b16 | .477 | .717 | .275 | .164 | 1.00 | .265 | .110 | .284 | .624 | .833 |
| CoOp ctx-init, biomedclip | .522 | .840 | .083 | .373 | .926 | .068 | .349 | .521 | .653 | .852 |
| CoOp + LP, siglip_b16 | .530 | .723 | .233 | .144 | .981 | .288 | .121 | .272 | .625 | .845 |
| CORAL, convnext_t | .440 | .700 | .924 | .687 | .685 | .227 | .405 | .421 | .920 | .937 |
| ERM, dinov2_lora | .171 | .683 | .933 | .443 | .815 | .583 | .413 | .582 | .806 | .963 |
| Fine-tune, swin_t | .328 | .702 | .910 | .483 | .907 | .114 | .318 | .617 | .872 | .954 |
| Fine-tune, NLL rule, convnext_t | .525 | .710 | .877 | .597 | .722 | .227 | .395 | .502 | .896 | .937 |
| Group DRO, convnext_t | .560 | .695 | .858 | .458 | .833 | .326 | .748 | .437 | .917 | .922 |
| Linear probe, medsiglip | .169 | .692 | .817 | .647 | .944 | .121 | .891 | .096 | .917 | .951 |
| MLP probe, dinov2_b14 | .380 | .650 | .864 | .701 | .722 | .053 | .451 | .119 | .879 | .898 |
| SWAD, convnext_t | .520 | .715 | .893 | .657 | .759 | .235 | .362 | .456 | .916 | .940 |
| Tip-Adapter, biomedclip | .245 | .362 | .675 | .353 | .574 | .553 | .015 | .977 | .392 | .406 |
| Tip-Adapter-F, biomedclip | .180 | .670 | .544 | .149 | .870 | .326 | .030 | .912 | .443 | .805 |
| WiSE-FT, siglip2_l16 | .323 | .675 | .881 | .537 | .852 | .197 | .228 | .617 | .834 | .928 |
| Zero-shot, medsiglip | .679 | .800 | .689 | .254 | .556 | .727 | .727 | .517 | .586 | .871 |

Table S12: ECE after TS per source, and the equal-source mean.

| Method | Paraguay | HYGD | FIVES | PAPILA | PALM | HRF | AFIO | Chákṣu | Eq-src |
|---|---|---|---|---|---|---|---|---|---|
| LOTUS | .046 | .084 | .049 | .058 | .094 | .301 | .203 | .091 | .119 |
| 16-shot probe, medsiglip | .075 | .103 | .075 | .321 | .124 | .175 | .301 | .060 | .168 |
| CoOp, siglip_b16 | .106 | .607 | .119 | .446 | .137 | .216 | .388 | .199 | .288 |
| CoOp ctx-init, biomedclip | .090 | .640 | .167 | .242 | .153 | .295 | .356 | .170 | .278 |
| CoOp + LP, siglip_b16 | .103 | .611 | .116 | .443 | .139 | .234 | .412 | .203 | .294 |
| CORAL, convnext_t | .055 | .444 | .184 | .385 | .181 | .291 | .249 | .206 | .256 |
| ERM, dinov2_lora | .044 | .215 | .305 | .288 | .113 | .278 | .234 | .113 | .211 |
| Fine-tune, swin_t | .043 | .480 | .250 | .411 | .170 | .331 | .333 | .239 | .288 |
| Fine-tune, NLL rule, convnext_t | .051 | .423 | .171 | .346 | .175 | .337 | .280 | .211 | .255 |
| Group DRO, convnext_t | .051 | .405 | .156 | .100 | .149 | .263 | .299 | .081 | .203 |
| Linear probe, medsiglip | .032 | .184 | .249 | .063 | .107 | .315 | .285 | .140 | .176 |
| MLP probe, dinov2_b14 | .046 | .617 | .179 | .328 | .130 | .303 | .313 | .215 | .274 |
| SWAD, convnext_t | .051 | .466 | .166 | .390 | .199 | .288 | .242 | .219 | .258 |
| Tip-Adapter, biomedclip | .070 | .315 | .065 | .362 | .184 | .137 | .085 | .170 | .174 |
| Tip-Adapter-F, biomedclip | .053 | .336 | .109 | .568 | .178 | .317 | .315 | .165 | .268 |
| WiSE-FT, siglip2_l16 | .063 | .442 | .250 | .484 | .083 | .338 | .294 | .084 | .279 |
| Zero-shot, medsiglip | .064 | .133 | .052 | .132 | .270 | .192 | .209 | .214 | .150 |

**Table S13.** LOTUS ECE after TS per source on the published split, three seeds pooled, with 95% bootstrap intervals over leakage groups (1,000 draws).

| Source | Images | Groups | ECE (TS) | 95% CI |
|---|---|---|---|---|
| Paraguay | 1,431 | 1,271 | .046 | [.037, .062] |
| HYGD | 548 | 183 | .084 | [.061, .123] |
| FIVES | 586 | 574 | .049 | [.042, .084] |
| PAPILA | 420 | 210 | .058 | [.050, .092] |
| PALM | 161 | 161 | .094 | [.079, .152] |
| HRF | 45 | 45 | .301 | [.196, .438] |
| AFIO | 129 | 65 | .203 | [.157, .297] |
| Chákṣu | 1,343 | 1,306 | .091 | [.078, .112] |

Table S14: Per-source accuracy, bAcc, macro-F1 (over classes present) and ECE after TS, three seeds. *Single-class source: bAcc equals that class's recall and macro-F1 is degenerate.

| Method | Source | $n$ | Classes | Acc. | bAcc | Macro-F1 | ECE (TS) |
|---|---|---|---|---|---|---|---|
| LOTUS | Paraguay | 1,431 | N/D | .920 | .919 | .936 | .046 |
| | HYGD* | 548 | G | .822 | .822 | .902 | .084 |
| | FIVES | 586 | N/D/G | .721 | .721 | .722 | .049 |
| | PAPILA | 420 | N/G | .802 | .664 | .712 | .058 |
| | PALM* | 161 | N | .836 | .836 | .911 | .094 |
| | HRF | 45 | N/D/G | .622 | .622 | .522 | .301 |
| | AFIO | 129 | N/D/G | .579 | .652 | .585 | .203 |
| | Chákṣu | 1,343 | N/G | .791 | .695 | .672 | .091 |
| 16-shot probe, MedSigLIP | Paraguay | 1,431 | N/D | .875 | .874 | .902 | .075 |
| | HYGD* | 548 | G | .756 | .756 | .858 | .103 |
| | FIVES | 586 | N/D/G | .670 | .668 | .656 | .075 |
| | PAPILA | 420 | N/G | .288 | .499 | .298 | .321 |
| | PALM* | 161 | N | .723 | .723 | .836 | .124 |
| | HRF | 45 | N/D/G | .711 | .711 | .713 | .175 |
| | AFIO | 129 | N/D/G | .413 | .548 | .391 | .301 |
| | Chákṣu | 1,343 | N/G | .619 | .675 | .572 | .060 |
| Linear probe, MedSigLIP | Paraguay | 1,431 | N/D | .934 | .934 | .940 | .032 |
| | HYGD* | 548 | G | .477 | .477 | .646 | .184 |
| | FIVES | 586 | N/D/G | .561 | .559 | .532 | .249 |
| | PAPILA | 420 | N/G | .726 | .493 | .512 | .063 |
| | PALM* | 161 | N | .907 | .907 | .951 | .107 |
| | HRF | 45 | N/D/G | .637 | .637 | .531 | .315 |
| | AFIO | 129 | N/D/G | .509 | .571 | .447 | .285 |
| | Chákṣu | 1,343 | N/G | .590 | .347 | .392 | .140 |
| Zero-shot, MedSigLIP | Paraguay | 1,431 | N/D | .730 | .729 | .809 | .064 |
| | HYGD* | 548 | G | .719 | .719 | .837 | .133 |
| | FIVES | 586 | N/D/G | .724 | .723 | .723 | .052 |
| | PAPILA | 420 | N/G | .683 | .622 | .638 | .132 |
| | PALM* | 161 | N | .379 | .379 | .550 | .270 |
| | HRF | 45 | N/D/G | .556 | .556 | .511 | .192 |
| | AFIO | 129 | N/D/G | .457 | .512 | .469 | .209 |
| | Chákṣu | 1,343 | N/G | .406 | .615 | .396 | .214 |
| Group DRO, ConvNeXt-T | Paraguay | 1,431 | N/D | .920 | .920 | .924 | .051 |
| | HYGD* | 548 | G | .316 | .316 | .454 | .405 |
| | FIVES | 586 | N/D/G | .704 | .704 | .703 | .156 |
| | PAPILA | 420 | N/G | .683 | .592 | .614 | .100 |
| | PALM* | 161 | N | .870 | .870 | .928 | .149 |
| | HRF | 45 | N/D/G | .600 | .600 | .538 | .263 |
| | AFIO | 129 | N/D/G | .465 | .539 | .457 | .299 |
| | Chákṣu | 1,343 | N/G | .686 | .417 | .451 | .081 |
| ERM, DINOv2 LoRA | Paraguay | 1,431 | N/D | .885 | .884 | .897 | .044 |
| | HYGD* | 548 | G | .505 | .505 | .669 | .215 |
| | FIVES | 586 | N/D/G | .597 | .596 | .558 | .305 |
| | PAPILA | 420 | N/G | .448 | .498 | .483 | .288 |
| | PALM* | 161 | N | .816 | .816 | .898 | .113 |
| | HRF | 45 | N/D/G | .607 | .607 | .557 | .278 |
| | AFIO | 129 | N/D/G | .543 | .614 | .542 | .234 |
| | Chákṣu | 1,343 | N/G | .615 | .419 | .481 | .113 |
| Fine-tune, Swin-T | Paraguay | 1,431 | N/D | .913 | .913 | .916 | .043 |
| | HYGD* | 548 | G | .237 | .237 | .380 | .480 |
| | FIVES | 586 | N/D/G | .647 | .647 | .630 | .250 |
| | PAPILA | 420 | N/G | .380 | .468 | .426 | .411 |
| | PALM* | 161 | N | .836 | .836 | .910 | .170 |
| | HRF | 45 | N/D/G | .600 | .600 | .524 | .331 |
| | AFIO | 129 | N/D/G | .416 | .501 | .369 | .333 |
| | Chákṣu | 1,343 | N/G | .487 | .298 | .365 | .239 |

*Reading*. DR is detected well by every method (AUROC 0.93–0.96). Glaucoma separates the methods: LOTUS reaches AUROC 0.926 on test against 0.79–0.87 for the baselines. On Chákṣu, fine-tuned and DG models detect almost no glaucoma (sensitivity 0.04–

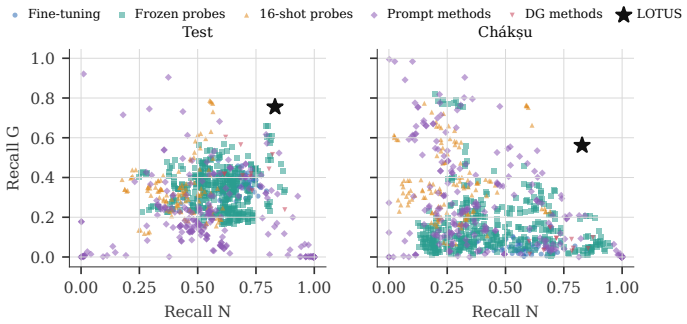



**Fig. S1.** Recall of normal against glaucoma images for every configuration. Few reach the upper right on either view: models either miss glaucoma or over-call it. LOTUS is among the most balanced on test; on Chákṣu it keeps normal recall high at the cost of glaucoma recall.

**Table S15.** Disease-level results, one disease against the rest, after TS, three seeds: sensitivity, specificity, positive predictive value and AUROC.

| Method | View | Disease | Sens. | Spec. | PPV | AUROC |
|---|---|---|---|---|---|---|
| LOTUS | Test | D | .905 | .931 | .842 | .961 |
| LOTUS | Test | G | .755 | .918 | .770 | .926 |
| LOTUS | Chákṣu | G | .561 | .866 | .404 | .841 |
| 16-shot probe, MedSigLIP | Test | D | .906 | .873 | .746 | .957 |
| 16-shot probe, MedSigLIP | Test | G | .783 | .767 | .550 | .870 |
| 16-shot probe, MedSigLIP | Chákṣu | G | .752 | .703 | .292 | .803 |
| Linear probe, MedSigLIP | Test | D | .896 | .887 | .762 | .940 |
| Linear probe, MedSigLIP | Test | G | .488 | .908 | .658 | .854 |
| Linear probe, MedSigLIP | Chákṣu | G | .011 | .999 | .600 | .768 |
| Zero-shot, MedSigLIP | Test | D | .850 | .913 | .798 | .944 |
| Zero-shot, MedSigLIP | Test | G | .682 | .771 | .521 | .811 |
| Zero-shot, MedSigLIP | Chákṣu | G | .904 | .351 | .184 | .757 |
| Group DRO, ConvNeXt-T | Test | D | .871 | .848 | .717 | .929 |
| Group DRO, ConvNeXt-T | Test | G | .443 | .918 | .658 | .846 |
| Group DRO, ConvNeXt-T | Chákṣu | G | .045 | .992 | .502 | .676 |
| ERM, DINOv2 LoRA | Test | D | .899 | .840 | .694 | .929 |
| ERM, DINOv2 LoRA | Test | G | .603 | .835 | .577 | .831 |
| ERM, DINOv2 LoRA | Chákṣu | G | .148 | .947 | .321 | .650 |
| Fine-tune, Swin-T | Test | D | .898 | .819 | .669 | .927 |
| Fine-tune, Swin-T | Test | G | .411 | .869 | .533 | .789 |
| Fine-tune, Swin-T | Chákṣu | G | .037 | .973 | .211 | .622 |

0.15); LOTUS has the highest specificity among methods that detect a meaningful share of glaucoma (0.866, sensitivity 0.561, PPV 0.404), whereas the initial design reached sensitivity 0.822 at specificity 0.688. Precision for glaucoma on Chákṣu stays low for every method, because only 14% of its images are glaucoma; the higher values of the linear probe and Group DRO (0.50–0.60) come with sensitivities below 0.05.

**Table S16.** Sensitivity to label scope and single-class sources: test bAcc on all sources, without the single-class HYGD and PALM, and on the four sources that assessed normal images for glaucoma (AFIO, FIVES, HRF, PAPILA; 1,180 images).

| Method | All test | Without single-class | Full-scope, multi-class |
|---|---|---|---|
| LOTUS | .830 | .794 | .728 |
| 16-shot probe, MedSigLIP | .748 | .756 | .608 |
| Linear probe, MedSigLIP | .728 | .730 | .623 |
| Zero-shot, MedSigLIP | .709 | .698 | .689 |
| Group DRO, ConvNeXt-T | .707 | .772 | .673 |
| ERM, DINOv2 LoRA | .707 | .752 | .605 |
| Fine-tune, Swin-T | .655 | .741 | .591 |

*Reading.* On the four sources that graded every class, LOTUS remains first (0.728), ahead of zero-shot (0.689) and the probes (0.608–0.623); every method is lower there than on the pooled test, which Paraguay inflates.

*Reading.* Paraguay is easy for every method, whereas AFIO and HRF are hard for all. D transfers well everywhere; the differences between methods come mostly from G and from N on G-heavy sources. Per-source ECE after a single global temperature remains high for every method on HRF and AFIO, small sets where a pooled ECE hides the error. Table S13 shows that their intervals are wide (HRF 0.196–0.438 with 45 images), so calibration on sets this small cannot be certified either way; a deployment site needs a local calibration set of a few hundred images (Table S21).

## S5 Reliability

Table S17: Selective prediction on test with thresholds fitted on Cal-B, raw/TS/FTS. Cov 80, Cov 90: thresholds for that coverage on Cal-B. Risk 10: SGR bound of 10%.

| | | Cov 80 | | | | Cov 90 | | Risk 10 | | Chákṣu Cov 80 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Method | | Cov. | Risk | G→N | D→N | Cov. | Risk | Cov. | Risk | Cov. | Risk |
| LOTUS | R | .691 | .083 | .054 | .035 | .827 | .114 | .716 | .087 | .442 | .064 |
| | T | .691 | .078 | .054 | .035 | .830 | .112 | .732 | .086 | .437 | .055 |
| | F | .690 | .078 | .054 | .035 | .833 | .111 | .737 | .087 | .439 | .051 |
| 16-shot probe, medsiglip | R | .757 | .196 | .012 | .017 | .883 | .245 | .354 | .060 | .528 | .307 |
| | T | .753 | .195 | .012 | .016 | .876 | .241 | .353 | .057 | .509 | .307 |
| | F | .752 | .195 | .012 | .016 | .876 | .241 | .357 | .058 | .504 | .307 |
| CoOp, siglip_b16 | R | .801 | .477 | .039 | .086 | .898 | .489 | .000 | – | .598 | .648 |
| | T | .797 | .476 | .038 | .089 | .898 | .488 | .000 | – | .572 | .649 |
| | F | .798 | .476 | .038 | .086 | .899 | .489 | .000 | – | .586 | .647 |
| CoOp ctx-init, biomedclip | R | .761 | .454 | .200 | .068 | .869 | .469 | .000 | – | .808 | .395 |
| | T | .745 | .453 | .187 | .065 | .854 | .468 | .000 | – | .811 | .398 |
| | F | .749 | .452 | .189 | .067 | .857 | .468 | .000 | – | .812 | .396 |
| CoOp + LP, siglip_b16 | R | .793 | .470 | .040 | .085 | .895 | .484 | .000 | – | .572 | .673 |
| | T | .796 | .470 | .040 | .090 | .898 | .484 | .000 | – | .551 | .675 |
| | F | .797 | .469 | .039 | .088 | .898 | .484 | .000 | – | .559 | .674 |

Table S17: (continued)

| Method | | Cov 80 Cov. | Cov 80 Risk | Cov 80 G→N | Cov 80 D→N | Cov 90 Cov. | Cov 90 Risk | Risk 10 Cov. | Risk 10 Risk | Chákṣu Cov 80 Cov. | Chákṣu Cov 80 Risk |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CORAL, convnext_t | R | .668 | .218 | .157 | .029 | .807 | .253 | .727 | .233 | .530 | .354 |
| | T | .655 | .212 | .144 | .028 | .789 | .245 | .706 | .224 | .525 | .353 |
| | F | .668 | .218 | .158 | .029 | .807 | .254 | .726 | .233 | .529 | .354 |
| ERM, dinov2_lora | R | .726 | .228 | .075 | .013 | .848 | .262 | .697 | .224 | .566 | .318 |
| | T | .715 | .223 | .067 | .012 | .829 | .255 | .689 | .221 | .552 | .314 |
| | F | .723 | .227 | .073 | .013 | .839 | .259 | .695 | .222 | .561 | .318 |
| Fine-tune, swin_t | R | .708 | .257 | .156 | .025 | .819 | .283 | .766 | .269 | .457 | .489 |
| | T | .694 | .250 | .147 | .024 | .803 | .276 | .752 | .263 | .444 | .484 |
| | F | .709 | .257 | .156 | .025 | .819 | .283 | .766 | .269 | .458 | .489 |
| Fine-tune, NLL rule, convnext_t | R | .706 | .219 | .141 | .033 | .824 | .253 | .765 | .237 | .497 | .423 |
| | T | .690 | .213 | .126 | .031 | .809 | .246 | .755 | .230 | .489 | .420 |
| | F | .707 | .220 | .143 | .033 | .826 | .254 | .766 | .238 | .497 | .423 |
| Group DRO, convnext_t | R | .734 | .180 | .086 | .042 | .834 | .208 | .661 | .162 | .569 | .249 |
| | T | .738 | .182 | .089 | .042 | .838 | .210 | .664 | .165 | .570 | .249 |
| | F | .751 | .187 | .097 | .044 | .850 | .215 | .668 | .167 | .572 | .249 |
| Linear probe, medsiglip | R | .602 | .127 | .057 | .022 | .750 | .161 | .794 | .176 | .323 | .271 |
| | T | .610 | .129 | .061 | .022 | .765 | .167 | .800 | .178 | .308 | .275 |
| | F | .620 | .132 | .067 | .023 | .771 | .169 | .811 | .183 | .299 | .276 |
| MLP probe, dinov2_b14 | R | .692 | .290 | .304 | .048 | .834 | .335 | .679 | .288 | .535 | .461 |
| | T | .672 | .281 | .281 | .045 | .812 | .326 | .666 | .278 | .524 | .459 |
| | F | .685 | .287 | .298 | .047 | .824 | .330 | .673 | .285 | .532 | .462 |
| SWAD, convnext_t | R | .708 | .225 | .172 | .032 | .832 | .257 | .820 | .253 | .478 | .401 |
| | T | .690 | .216 | .156 | .030 | .813 | .246 | .804 | .245 | .471 | .399 |
| | F | .704 | .223 | .167 | .032 | .825 | .252 | .816 | .250 | .475 | .399 |
| Tip-Adapter, biomedclip | R | .798 | .549 | .095 | .153 | .897 | .554 | .000 | – | .794 | .495 |
| | T | .757 | .547 | .089 | .113 | .874 | .549 | .000 | – | .930 | .490 |
| | F | .762 | .546 | .091 | .116 | .872 | .548 | .000 | – | .923 | .490 |
| Tip-Adapter-F, biomedclip | R | .769 | .526 | .079 | .045 | .882 | .526 | .000 | – | .810 | .409 |
| | T | .771 | .524 | .077 | .046 | .887 | .525 | .028 | .446 | .828 | .416 |
| | F | .770 | .524 | .077 | .046 | .887 | .526 | .030 | .458 | .828 | .416 |
| WiSE-FT, siglip2_l16 | R | .659 | .270 | .100 | .028 | .804 | .306 | .679 | .276 | .430 | .286 |
| | T | .641 | .260 | .088 | .025 | .780 | .295 | .661 | .268 | .417 | .280 |
| | F | .654 | .267 | .097 | .027 | .798 | .303 | .673 | .274 | .426 | .283 |
| Zero-shot, medsiglip | R | .727 | .230 | .122 | .052 | .868 | .270 | .190 | .040 | .745 | .581 |
| | T | .711 | .224 | .107 | .052 | .861 | .267 | .182 | .033 | .689 | .576 |
| | F | .711 | .224 | .107 | .052 | .861 | .267 | .182 | .033 | .689 | .576 |

Table S18: Conformal sets on test, quantiles from Cal-B. Coverage/mean size at nominal 0.90; coverage at nominal 0.95.

| Method | | LAC .90 | APS .90 | Class-LAC .90 | LAC .95 | APS .95 | Class-LAC .95 | Chákṣu LAC .90 |
|---|---|---|---|---|---|---|---|---|
| LOTUS | R | .873/1.11 | .939/1.72 | .847/1.09 | .954 | .952 | .942 | .890/1.25 |
| | T | .883/1.14 | .955/1.88 | .848/1.10 | .956 | .957 | .941 | .902/1.28 |
| | F | .883/1.14 | .928/1.70 | .849/1.09 | .955 | .940 | .941 | .900/1.27 |
| 16-shot probe, medsiglip | R | .913/1.61 | .923/1.99 | .823/1.40 | .966 | .994 | .919 | .930/2.09 |
| | T | .912/1.61 | .922/1.98 | .817/1.39 | .967 | .996 | .916 | .929/2.08 |
| | F | .909/1.60 | .925/2.01 | .822/1.39 | .966 | .996 | .922 | .927/2.09 |
| CoOp, siglip_b16 | R | .761/1.80 | .866/2.24 | .761/1.83 | .843 | .947 | .856 | .817/2.26 |
| | T | .760/1.80 | .862/2.24 | .760/1.83 | .839 | .944 | .858 | .811/2.23 |
| | F | .760/1.79 | .852/2.20 | .773/1.85 | .839 | .986 | .858 | .802/2.18 |
| CoOp ctx-init, biomedclip | R | .777/1.89 | .762/1.91 | .784/1.95 | .846 | .878 | .845 | .854/1.77 |
| | T | .766/1.84 | .760/1.90 | .783/1.94 | .835 | .878 | .846 | .856/1.75 |
| | F | .759/1.81 | .750/1.87 | .782/1.93 | .825 | .875 | .850 | .858/1.75 |
| CoOp + LP, siglip_b16 | R | .755/1.78 | .809/2.01 | .761/1.81 | .839 | .949 | .847 | .802/2.27 |
| | T | .753/1.77 | .808/2.00 | .761/1.81 | .837 | .983 | .849 | .792/2.22 |
| | F | .755/1.77 | .795/1.96 | .773/1.84 | .832 | .985 | .853 | .785/2.18 |
| CORAL, convnext_t | R | .716/1.09 | .815/1.68 | .668/1.13 | .780 | .823 | .754 | .635/1.15 |
| | T | .710/1.07 | .807/1.60 | .663/1.10 | .773 | .819 | .749 | .637/1.15 |
| | F | .713/1.08 | .796/1.56 | .666/1.11 | .769 | .826 | .741 | .637/1.16 |
| ERM, dinov2_lora | R | .728/1.10 | .870/1.72 | .697/1.13 | .801 | .885 | .765 | .684/1.17 |
| | T | .722/1.08 | .870/1.72 | .689/1.11 | .796 | .884 | .758 | .679/1.15 |
| | F | .723/1.08 | .861/1.68 | .692/1.11 | .793 | .882 | .757 | .679/1.15 |
| Fine-tune, swin_t | R | .682/1.08 | .794/1.65 | .635/1.05 | .737 | .803 | .720 | .566/1.18 |
| | T | .676/1.06 | .786/1.59 | .631/1.03 | .732 | .798 | .715 | .559/1.16 |
| | F | .680/1.07 | .784/1.67 | .634/1.04 | .730 | .806 | .714 | .562/1.17 |
| Fine-tune, NLL rule, convnext_t | R | .712/1.08 | .829/1.73 | .679/1.08 | .771 | .838 | .769 | .604/1.18 |
| | T | .708/1.06 | .828/1.71 | .674/1.06 | .764 | .836 | .762 | .604/1.18 |
| | F | .711/1.07 | .806/1.58 | .679/1.07 | .763 | .829 | .757 | .602/1.18 |
| Group DRO, convnext_t | R | .767/1.13 | .844/1.65 | .733/1.09 | .810 | .854 | .814 | .787/1.29 |
| | T | .769/1.13 | .844/1.65 | .733/1.09 | .811 | .854 | .815 | .788/1.29 |
| | F | .771/1.14 | .838/1.65 | .735/1.10 | .811 | .855 | .810 | .790/1.30 |
| Linear probe, medsiglip | R | .743/0.99 | .883/1.68 | .682/0.92 | .857 | .893 | .765 | .631/1.09 |
| | T | .751/1.01 | .888/1.72 | .687/0.94 | .863 | .897 | .765 | .626/1.08 |
| | F | .754/1.01 | .855/1.55 | .691/0.95 | .857 | .876 | .760 | .626/1.08 |
| MLP probe, dinov2_b14 | R | .658/1.11 | .882/2.28 | .637/1.13 | .746 | .931 | .703 | .576/1.17 |
| | T | .651/1.09 | .798/1.79 | .629/1.11 | .739 | .807 | .695 | .573/1.16 |
| | F | .652/1.09 | .777/1.67 | .633/1.12 | .738 | .798 | .696 | .573/1.17 |
| SWAD, convnext_t | R | .698/1.04 | .827/1.73 | .668/1.05 | .761 | .834 | .766 | .562/1.09 |
| | T | .690/1.01 | .822/1.69 | .662/1.02 | .753 | .831 | .758 | .558/1.08 |
| | F | .693/1.02 | .802/1.54 | .666/1.03 | .753 | .825 | .750 | .560/1.09 |
| Tip-Adapter, biomedclip | R | .947/2.50 | .989/2.92 | .785/2.00 | .979 | .989 | .866 | .994/2.27 |

Table S18: (continued)

| Method | | LAC .90 | APS .90 | Class-LAC .90 | LAC .95 | APS .95 | Class-LAC .95 | Chákṣu LAC .90 |
|---|---|---|---|---|---|---|---|---|
| | T | .970/2.54 | .993/2.95 | .756/1.85 | .988 | .993 | .845 | 1.00/2.24 |
| | F | .961/2.52 | .992/2.94 | .764/1.90 | .984 | .992 | .840 | .997/2.25 |
| Tip-Adapter-F, biomedclip | R | .819/2.09 | .989/2.92 | .764/1.88 | .889 | .989 | .838 | .897/2.09 |
| | T | .825/2.09 | .992/2.94 | .766/1.86 | .899 | .992 | .841 | .901/2.10 |
| | F | .817/2.08 | .948/2.70 | .763/1.87 | .889 | .990 | .837 | .898/2.10 |
| WiSE-FT, siglip2_l16 | R | .676/1.12 | .805/1.78 | .633/1.13 | .738 | .813 | .707 | .717/1.23 |
| | T | .668/1.10 | .804/1.77 | .628/1.11 | .732 | .812 | .700 | .707/1.20 |
| | F | .671/1.11 | .790/1.70 | .631/1.12 | .729 | .811 | .701 | .709/1.21 |
| Zero-shot, medsiglip | R | .923/1.74 | .896/1.80 | .912/1.87 | .981 | .996 | .967 | .852/1.91 |
| | T | .924/1.76 | .920/1.89 | .916/1.90 | .982 | .994 | .968 | .842/1.90 |
| | F | .924/1.76 | .920/1.89 | .916/1.90 | .982 | .994 | .968 | .842/1.90 |

*Reading.* Methods whose sets stay near nominal coverage (16-shot, zero-shot) do so with mean set sizes of 1.6 to 1.8 out of three classes, which is of little use. Methods with compact sets lose coverage. TS keeps the ranking of confidence, so it barely changes coverage or risk at a fixed threshold.

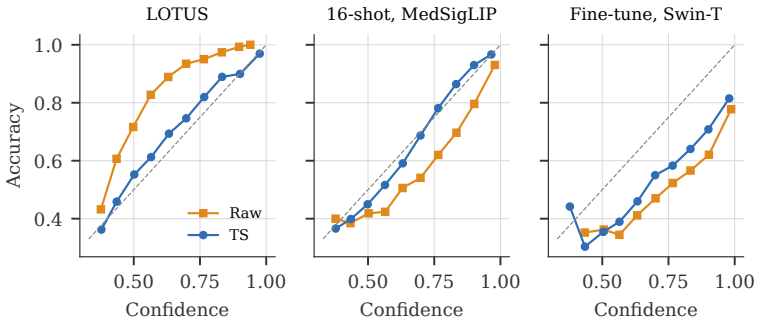



**Fig. S2.** Test reliability on the published split, seeds pooled, bins with at least 30 images. Raw LOTUS is underconfident and TS brings it near the diagonal; fine-tuned Swin-T stays overconfident after TS.

*Reading.* LOTUS has the lowest area under the risk–coverage curve of every method, on test (0.062; next: linear probe 0.110) and on Chákṣu (0.081; next: Group DRO 0.269). The Cal-B threshold lands at 69% coverage instead of 80%, and on Chákṣu at 44%: a threshold fitted on development data does not keep its coverage, and risk alone rewards models that call everything normal.

*Reading.* On the other half of Cal-B, the promises hold almost exactly; on test and Chákṣu they fail. The failure therefore comes from the new sources, not from a small calibration set.

*Reading.* The pooled risk target held (0.082), but per source it held on four of seven test sources and failed on AFIO, HRF, PALM and Chákṣu. A pooled guarantee is not a per-site guarantee, and neither is distribution-free under source shift; a deployment site must check its own promises (Table S37).

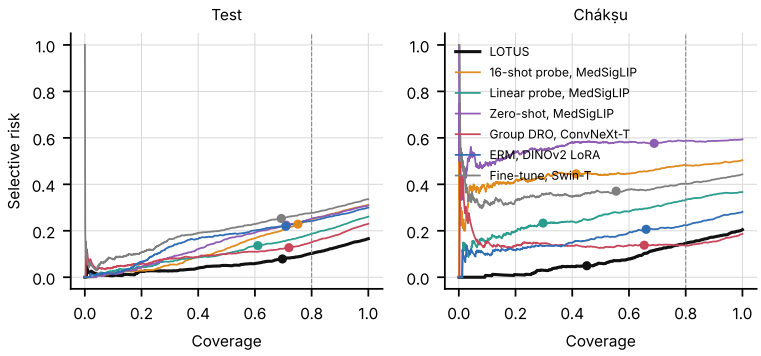



**Fig. S3.** Risk–coverage curves after TS on the published split (seed 0); dots mark the threshold fitted for 80% coverage on Cal-B. On Chákṣu, where 86% of images are normal, methods that rarely call glaucoma reach low risk while missing most disease.

**Table S19.** Is the failure caused by too few calibration images? Cal-B is split by leakage group 100 times; thresholds fit on one half are read on the other half (same sources), on test and on Chákṣu (TS).

| | Coverage at Cov 80 | | | Risk at Risk 10 | | | LAC .90 coverage | | |
|---|---|---|---|---|---|---|---|---|---|
| Method | Other half | Test | Chákṣu | Other half | Test | Chákṣu | Other half | Test | Chákṣu |
| 16-shot probe, medsiglip | .799 | .754 | .507 | .054 | .041 | .132 | .900 | .911 | .926 |
| CoOp, siglip_b16 | .799 | .797 | .573 | – | – | – | .901 | .760 | .811 |
| CoOp ctx-init, biomedclip | .800 | .744 | .810 | .134 | .257 | .294 | .900 | .767 | .856 |
| CoOp + LP, siglip_b16 | .799 | .795 | .548 | – | – | – | .901 | .756 | .796 |
| CORAL, convnext_t | .797 | .650 | .522 | .068 | .209 | .357 | .901 | .710 | .632 |
| ERM, dinov2_lora | .799 | .714 | .550 | .068 | .210 | .289 | .900 | .722 | .682 |
| Fine-tune, swin_t | .800 | .695 | .446 | .069 | .251 | .487 | .901 | .675 | .554 |
| Fine-tune, NLL rule, convnext_t | .798 | .688 | .487 | .068 | .213 | .424 | .902 | .709 | .605 |
| Group DRO, convnext_t | .797 | .737 | .569 | .064 | .150 | .248 | .900 | .771 | .789 |
| Linear probe, medsiglip | .800 | .610 | .308 | .070 | .161 | .318 | .901 | .752 | .632 |
| MLP probe, dinov2_b14 | .801 | .673 | .526 | .069 | .258 | .452 | .901 | .651 | .573 |
| SWAD, convnext_t | .798 | .689 | .470 | .070 | .230 | .423 | .901 | .691 | .560 |
| Tip-Adapter, biomedclip | .797 | .757 | .928 | – | – | – | .900 | .970 | 1.00 |
| Tip-Adapter-F, biomedclip | .801 | .772 | .829 | .084 | .436 | .642 | .900 | .827 | .904 |
| WiSE-FT, siglip2_l16 | .799 | .641 | .420 | .068 | .258 | .281 | .900 | .668 | .708 |
| Zero-shot, medsiglip | .799 | .717 | .696 | .045 | .027 | .432 | .902 | .925 | .844 |

**Table S20.** LOTUS's promises per unseen source on the published split (TS; thresholds fitted on Cal-B): coverage and risk at the thresholds for 80% and 90% coverage and at the SGR threshold for risk ≤ 10%, and whether that risk target held.

| | Cov 80 | | Cov 90 | | Risk 10 | | |
|---|---|---|---|---|---|---|---|
| Source | Cov. | Risk | Cov. | Risk | Cov. | Risk | Held |
| AFIO | .450 | .385 | .646 | .432 | .509 | .386 | no |
| FIVES | .573 | .128 | .749 | .192 | .620 | .135 | no |
| HRF | .859 | .345 | .948 | .367 | .889 | .358 | no |
| HYGD | .581 | .096 | .793 | .122 | .642 | .105 | no |
| PALM | .584 | .074 | .758 | .096 | .650 | .076 | yes |
| PAPILA | .623 | .071 | .796 | .131 | .671 | .088 | yes |
| Paraguay | .829 | .038 | .909 | .050 | .854 | .040 | yes |
| Chákṣu | .437 | .055 | .696 | .123 | .511 | .067 | yes |

Table S21: Calibration budget: fitted $T$ on all of Cal-A, the oracle $T$ fitted on test, and test NLL after TS with natural random subsets of Cal-A (median of 100 draws). $T = 1000$ marks the upper bound of the temperature search (zero-shot BiomedCLIP, whose calibrated output is then uniform, NLL = $\ln 3 \approx 1.10$).

| Setting | Model | $T$ | $T$ oracle | NLL, 50 | NLL, 200 | NLL, 1000 | NLL, all |
|---|---|---|---|---|---|---|---|
| 16-shot probe | biomedclip | 1.59 | 2.88 | 1.05 | 1.05 | 1.05 | 1.05 |
| 16-shot probe | clip_b16 | 1.80 | 3.28 | 1.08 | 1.08 | 1.07 | 1.07 |
| 16-shot probe | dinov2_b14 | 2.80 | 4.47 | .997 | .994 | .993 | .993 |
| 16-shot probe | resnet50 | 3.75 | 2.61 | .979 | .976 | .977 | .977 |
| 16-shot probe | siglip2_l16 | 1.57 | 1.75 | .929 | .926 | .926 | .926 |
| CoOp | biomedclip | 2.07 | 6.48 | 1.23 | 1.22 | 1.23 | 1.23 |
| CoOp | clip_b16 | 1.43 | 4.17 | 1.21 | 1.20 | 1.19 | 1.20 |
| CoOp | siglip2_l16 | .780 | .893 | 1.07 | 1.07 | 1.07 | 1.07 |
| CoOp ctx-init | biomedclip | 2.22 | 6.34 | 1.19 | 1.17 | 1.18 | 1.18 |
| CoOp ctx-init | clip_b16 | 1.25 | 3.88 | 1.20 | 1.19 | 1.19 | 1.19 |
| CoOp ctx-init | siglip2_l16 | 1.32 | 3.97 | 1.20 | 1.20 | 1.19 | 1.19 |
| CoOp + LP | biomedclip | 2.17 | 6.89 | 1.24 | 1.23 | 1.24 | 1.24 |
| CoOp + LP | clip_b16 | 1.51 | 4.39 | 1.21 | 1.20 | 1.19 | 1.19 |
| CoOp + LP | siglip2_l16 | .358 | .385 | .898 | .896 | .896 | .895 |
| Fine-tune | resnet50 | 1.52 | 3.65 | 1.02 | 1.01 | 1.01 | 1.01 |
| Fine-tune, NLL rule | resnet50 | 1.69 | 4.68 | 1.13 | 1.11 | 1.12 | 1.12 |
| Linear probe | biomedclip | .866 | 1.79 | 1.06 | 1.04 | 1.04 | 1.04 |
| Linear probe | clip_b16 | .889 | 1.83 | .971 | .944 | .951 | .950 |
| Linear probe | dinov2_b14 | .999 | 2.04 | .926 | .891 | .904 | .901 |
| Linear probe | resnet50 | 1.16 | 1.87 | .924 | .911 | .911 | .910 |
| Linear probe | siglip2_l16 | .999 | 2.22 | 1.01 | .989 | .993 | .993 |
| Multiple choice | Fundus-R1-7B | 1.87 | 2.24 | .821 | .816 | .817 | .817 |
| Multiple choice | Qwen2.5-VL-7B-Instruct | 1.78 | 3.39 | 1.07 | 1.08 | 1.08 | 1.08 |
| Multiple choice | llava-med-v1.5-mistral-7b-hf | .725 | .483 | 1.07 | 1.07 | 1.07 | 1.07 |
| Multiple choice | llava-onevision-qwen2-7b-ov-hf | 1.96 | 11.32 | 1.30 | 1.28 | 1.28 | 1.28 |
| MLP probe | biomedclip | 1.74 | 13.47 | 1.90 | 1.85 | 1.85 | 1.85 |
| MLP probe | clip_b16 | 2.36 | 16.03 | 1.99 | 1.97 | 1.95 | 1.94 |
| MLP probe | dinov2_b14 | 2.09 | 7.61 | 1.32 | 1.25 | 1.26 | 1.26 |
| MLP probe | siglip2_l16 | 1.91 | 11.20 | 1.78 | 1.75 | 1.74 | 1.73 |
| Zero-shot | biomedclip | 1000.00 | 137.44 | 1.10 | 1.10 | 1.10 | 1.10 |
| Zero-shot | clip_b16 | .785 | 2.12 | 1.15 | 1.15 | 1.15 | 1.15 |
| Zero-shot | siglip2_l16 | .604 | 1.69 | 1.17 | 1.18 | 1.17 | 1.17 |

*Reading.* About 200 calibration images give nearly the NLL of the full set, yet the fitted $T$ stays far from the test oracle for several models: more calibration data from the same sources does not close the shift.

**Table S22.** Trivial predictors, the floor for every metric.

| View | Predictor | Acc. | bAcc | Eq-src | ECE | NLL |
|---|---|---|---|---|---|---|
| Test | uniform | .445 | .333 | .429 | .112 | 1.10 |
| Test | train_prior | .445 | .333 | .429 | .063 | 1.30 |
| Test | always_N | .445 | .333 | .429 | – | – |
| Test | always_D | .288 | .333 | .214 | – | – |
| Test | always_G | .267 | .333 | .357 | – | – |
| Chákṣu | uniform | .861 | .500 | .500 | .527 | 1.10 |
| Chákṣu | train_prior | .861 | .500 | .500 | .353 | .978 |
| Chákṣu | always_N | .861 | .500 | .500 | – | – |
| Chákṣu | always_D | .000 | .000 | .000 | – | – |
| Chákṣu | always_G | .139 | .500 | .500 | – | – |

**Table S23.** Spearman correlation between validation and test bAcc across configurations within each setting. Settings with at most two configurations (the four DG methods and the NLL rule, 9 configurations) are not listed separately but are counted in the last row.

| Setting | Configurations | Spearman |
|---|---|---|
| 16-shot probe | 130 | .374 |
| CoOp | 70 | .800 |
| CoOp ctx-init | 70 | .927 |
| CoOp + LP | 50 | .393 |
| Fine-tune | 65 | .053 |
| Linear probe | 390 | .344 |
| MLP probe | 11 | .527 |
| Tip-Adapter | 3 | .500 |
| Tip-Adapter (acc. rule) | 3 | .500 |
| Tip-Adapter-F | 3 | −.500 |
| Tip-Adapter-F (acc. rule) | 3 | −.500 |
| WiSE-FT | 4 | -1.00 |
| Zero-shot | 36 | .831 |
| All configurations | 847 | .781 |

*Reading.* A class-prior predictor reaches ECE 0.063 with chance bAcc, so low ECE alone says nothing about usefulness. The high overall rank correlation (0.781 here, over 847 rows that also count the accuracy-rule Tip-Adapter variants; 0.786 over the 816 configurations of the main paper) mostly reflects differences between families; within fine-tuning it is 0.053.

## S6 Robustness Checks

**Table S24.** Fifty random 7/7 source rotations (published split excluded). Probes use class-weighted cross-entropy; arrows go from internal estimates on held-back Cal-B to unseen test sources, after TS.

| Method | Val | Test | 5–95% | Eq-src | Rec. G | ECE raw | ECE TS | Risk at Cov 80 | LAC .90 | Coverage .80 |
|---|---|---|---|---|---|---|---|---|---|---|
| LOTUS | .876 | .788 | .721–.842 | .766 | .708 | .161 | .034 | .065→.101 | .900→.848 | .795→.722 |
| medsiglip, full | .922 | .745 | .621–.827 | .725 | .637 | .095 | .084 | .042→.134 | .901→.769 | .795→.668 |
| medsiglip, 16-shot | .814 | .716 | .626–.817 | .713 | .723 | .123 | .056 | .164→.217 | .899→.868 | .803→.778 |
| dinov2_b14, full | .892 | .655 | .529–.765 | .650 | .531 | .123 | .118 | .077→.221 | .899→.735 | .794→.659 |
| dinov2_b14, 16-shot | .713 | .552 | .432–.659 | .571 | .505 | .192 | .073 | .296→.383 | .900→.860 | .801→.779 |
| siglip2_l16, full | .876 | .631 | .549–.731 | .617 | .549 | .155 | .141 | .102→.280 | .900→.728 | .804→.680 |
| siglip2_l16, 16-shot | .722 | .578 | .476–.683 | .562 | .586 | .160 | .080 | .292→.372 | .901→.871 | .796→.751 |
| convnext_t, full | .860 | .593 | .483–.700 | .609 | .426 | .149 | .130 | .112→.287 | .901→.730 | .800→.684 |
| convnext_t, 16-shot | .679 | .494 | .379–.604 | .504 | .456 | .223 | .091 | .352→.463 | .900→.840 | .795→.771 |

*Reading.* Validation exceeded test in 98% of the 400 probe rows, risk exceeded its internal estimate in 89%, and LAC fell below its internal coverage in 85% (338 of 400); for LOTUS the shares were 98%, 86% and 92%. Ranking is not stable either: the 16-shot probe, best on the published split among representatives, is below full training on every encoder over the rotations.

**Table S25.** Full factorial ablation over the 50 rotations: every combination of bag, between-site projection and text, and text alone; TS, means over three seeds. Last two columns: LOTUS minus the row in bAcc, and rotations where LOTUS is higher.

| Configuration | bAcc | Macro-F1 | AUROC | Rec. G | False G | ECE | NLL | AURC | Risk | Chákṣu | ΔbAcc | Higher |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **LOTUS** (bag + text) | .789 | .739 | .931 | .709 | .070 | .034 | .507 | .071 | .101 | .694 | – | – |
| single head + text | .786 | .737 | .930 | .695 | .069 | .035 | .509 | .071 | .099 | .688 | +0.003 | 35/50 |
| bag only | .779 | .720 | .928 | .725 | .088 | .050 | .566 | .090 | .106 | .598 | +0.010 | 27/50 |
| + projection (initial design) | .771 | .718 | .921 | .666 | .081 | .033 | .526 | .075 | .117 | .668 | +0.018 | 45/50 |
| single head | .770 | .713 | .924 | .699 | .086 | .053 | .577 | .093 | .106 | .590 | +0.019 | 35/50 |
| single head + projection + text | .768 | .715 | .920 | .656 | .080 | .033 | .530 | .076 | .116 | .663 | +0.021 | 47/50 |
| bag + projection | .754 | .694 | .911 | .657 | .099 | .055 | .600 | .094 | .125 | .624 | +0.035 | 48/50 |
| single head + projection | .745 | .686 | .904 | .637 | .101 | .058 | .618 | .100 | .130 | .612 | +0.044 | 49/50 |
| text only | .713 | .653 | .886 | .614 | .131 | .022 | .611 | .109 | .190 | .594 | +0.076 | 50/50 |

*Reading.* Every arm with the projection is below the same arm without it (bag + text: 0.789 against 0.771; single head + text: 0.786 against 0.768; bag: 0.779 against 0.754; single head: 0.770 against 0.745), so the projection's cost is consistent rather than an interaction. Text lowers ECE and NLL in every combination; the bag adds little once text is present (0.003) but more without it (0.009 for the single head).

**Table S26.** Three training seeds per rotation (rotations where all three finished; the probe uses class-weighted cross-entropy as in Table S24). bAcc: mean ± SD of seed-averaged bAcc across rotations; seed SD: SD across seeds within a rotation, averaged. Last columns: LOTUS minus the row, mean ± SD over rotations, and rotations where LOTUS has higher bAcc and lower ECE.

| Configuration | Rot. | bAcc | Seed SD | ECE | Rec. G | ΔbAcc | Higher | Lower ECE |
|---|---|---|---|---|---|---|---|---|
| **LOTUS** | 50 | .789 ± .039 | .004 | .034 | .709 | – | – | – |
| + projection (initial design) | 50 | .771 ± .039 | .003 | .033 | .666 | +0.018 ± 0.015 | 45/50 | 23/50 |
| single head + text | 50 | .786 ± .041 | .005 | .035 | .695 | +0.003 ± 0.007 | 35/50 | 33/50 |
| bag only | 50 | .779 ± .050 | .007 | .050 | .725 | +0.010 ± 0.028 | 27/50 | 34/50 |
| text only | 50 | .713 ± .036 | .000 | .022 | .614 | +0.076 ± 0.027 | 50/50 | 11/50 |
| MedSigLIP linear probe | 50 | .746 ± .069 | .010 | .084 | .642 | +0.043 ± 0.049 | 43/50 | 47/50 |

*Reading.* Seed variation is an order of magnitude smaller than variation between source partitions, so single-seed rotations measure the partition effect. Against the probe, LOTUS has higher bAcc on 43 and lower ECE on 47 of 50 rotations; text only is better calibrated (lower ECE than LOTUS on 39 of 50) but far less accurate.

*Reading.* Fitted on all development data ($k = 14$), the between-site directions hold 79–87% of every class-mean difference; for the individual heads ($k = 11$–$13$) they hold 55–87%, against 15–23% for a random subspace of equal size. Sites differ in severity and case mix within each label, so their within-class differences are themselves disease directions; projecting them out removes signal, which explains the accuracy loss over the rotations.

*Reading.* The source is almost perfectly decodable from every representation, frozen or fine-tuned (0.93–1.00), even within one class. Fine-tuning does not add site information,

**Table S27.** Why projection hurts. Share of the norm of each class-mean difference that lies in the removed between-site subspace ($k$ directions), on the development data used by each head, against a random subspace of the same size; and alignment between the two glaucoma sites' difference and the N–G direction.

| Source left out | $k$ | N–D | N–G | D–G | N–G, random | \| cos \| G-site, N–G |
|---|---|---|---|---|---|---|
| none (all) | 14 | .868 | .788 | .811 | .187 | .098 |
| Bangladesh Eye | 11 | .839 | .548 | .602 | .225 | – |
| OIA-DDR | 12 | .830 | .839 | .817 | .208 | .073 |
| DeepDRiD | 12 | .850 | .776 | .762 | .208 | .083 |
| GRAPE | 13 | .866 | .819 | .836 | .147 | – |
| IDRiD | 12 | .818 | .784 | .793 | .209 | .100 |
| Jichi | 12 | .832 | .726 | .747 | .218 | .048 |
| SUSTech-SYSU | 12 | .791 | .785 | .782 | .207 | .115 |

**Table S28.** How much of the source can be read from each representation. Five-fold cross-validated balanced accuracy of a logistic probe (C = 0.05) predicting the disease, the source, or the source within normal or DR images only, on development (validation and calibration) rows and on test rows. Chance for the source is 1/6 or 1/7.

| Features | Rows | $n$ | Disease | Source | Source within N | Source within D |
|---|---|---|---|---|---|---|
| Fine-tuned Swin-T | development | 5,951 | .891 | .951 | .963 | .931 |
| Fine-tuned Swin-T | test | 3,320 | .908 | .977 | .996 | .996 |
| Group DRO ConvNeXt-T | development | 5,951 | .867 | .930 | .933 | .909 |
| Group DRO ConvNeXt-T | test | 3,320 | .906 | .980 | .994 | .944 |
| ERM DINOv2 LoRA | development | 5,951 | .873 | .946 | .967 | .914 |
| ERM DINOv2 LoRA | test | 3,320 | .911 | .986 | .998 | .995 |
| Frozen MedSigLIP (pooled) | development | 5,951 | .895 | .961 | .949 | .949 |
| Frozen MedSigLIP (pooled) | test | 3,320 | .931 | .998 | .999 | .986 |
| Frozen MedSigLIP tokens (LOTUS input) | development | 5,951 | .890 | .962 | .958 | .959 |
| Frozen MedSigLIP tokens (LOTUS input) | test | 3,320 | .924 | .998 | .999 | .986 |
| Tokens after projection (initial design) | development | 5,951 | .855 | .354 | .318 | .322 |
| Tokens after projection (initial design) | test | 3,320 | .924 | .998 | .999 | .984 |

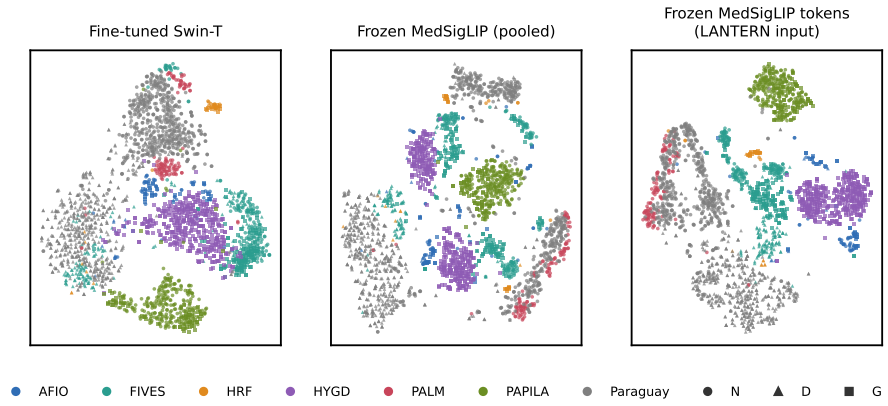



**Fig. S4.** t-SNE of test images (2,500 sampled), coloured by source, marker by class, for a fine-tuned encoder and the two frozen MedSigLIP representations. Every representation clusters by source.

so its failure lies in what its classifier relies on, not in what its features contain; frozen features carry the same site signal, and a low-capacity head simply exploits it less. The between-site projection removes site information for the development sources (0.962 to 0.354) but not for unseen ones (0.998), while it costs disease signal (0.890 to 0.855): it removes the known sites' directions, and new sites differ along new ones.

**Table S29.** Glaucoma recall over the 50 rotations by the number of development sources that contain glaucoma.

| Method | G sources in development | Rotations | Recall G | Test bAcc |
|---|---|---|---|---|
| LOTUS | 2 | 8 | .614 | .777 |
| LOTUS | 3 | 18 | .718 | .793 |
| LOTUS | 4 | 17 | .723 | .789 |
| LOTUS | 5 | 6 | .756 | .788 |
| LOTUS | 6 | 1 | .777 | .821 |
| Initial design | 2 | 8 | .555 | .761 |
| Initial design | 3 | 18 | .671 | .774 |
| Initial design | 4 | 17 | .702 | .774 |
| Initial design | 5 | 6 | .684 | .762 |
| Initial design | 6 | 1 | .748 | .801 |
| MedSigLIP LP | 2 | 8 | .468 | .676 |
| MedSigLIP LP | 3 | 18 | .639 | .742 |
| MedSigLIP LP | 4 | 17 | .674 | .766 |
| MedSigLIP LP | 5 | 6 | .737 | .772 |
| MedSigLIP LP | 6 | 1 | .940 | .878 |

*Reading.* Glaucoma recall grows with the number of glaucoma sources on the development side, so the weakness is one of site diversity, which more sources can address. LOTUS stays ahead of the linear probe at every diversity level observed in more than one rotation (two sources: 0.777 against 0.676; five: 0.788 against 0.772), whereas the initial design fell behind the probe with five sources (0.762 against 0.772).

*Reading.* Region hold-out is harsher than the rotations for some regions: holding out East Asia removes GRAPE, leaving one glaucoma development source. LOTUS has the highest mean bAcc (0.822), clearly ahead on South Asia (0.751 against 0.678) and the Middle East, within 0.013 of the probe elsewhere, and the best Chákṣu bAcc in every hold-out; its calibration is not better than the probe's here (mean ECE 0.061 against 0.058).

*Reading.* N–D and D–G stay near 0.90–0.95 while N–G is 0.64–0.75, so glaucoma is a ranking failure, not a threshold failure. Within-source learning recovers much of it (Chákṣu gap +0.372), so the labels are learnable and the loss is tied to the shift. AFIO stays low even within source.

*Reading.* More sources usually help on test (MedSigLIP 0.594 to 0.817 from one to seven sources) but not on Chákṣu (0.534 to 0.350), and more labels from the same sources give little after about 64 per class. Shift is not solved by more of the same data.

**Table S30.** Leave-one-region-out: all non-OOD sources of one region form the test side, the rest the development side (80/10/10 group split), three seeds. Eq-src: mean per-source bAcc; Chákṣu is read for every split.

| Region held out | Test sources | Images | Method | bAcc | Eq-src | Rec. G | ECE | Chákṣu |
|---|---|---|---|---|---|---|---|---|
| East Asia | OIA-DDR, DeepDRiD, FIVES, GRAPE, Jichi, PALM, SUSTech-SYSU | 26,487 | **LOTUS** | .810 | .820 | .822 | .058 | .728 |
| | | | Initial design | .773 | .777 | .743 | .047 | .650 |
| | | | MedSigLIP LP | .810 | .822 | .815 | .055 | .559 |
| Europe | HRF, PAPILA | 465 | **LOTUS** | .726 | .646 | .448 | .051 | .705 |
| | | | Initial design | .740 | .662 | .657 | .057 | .680 |
| | | | MedSigLIP LP | .739 | .662 | .497 | .055 | .562 |
| Latin America | Paraguay | 1,431 | **LOTUS** | .913 | .913 | – | .044 | .726 |
| | | | Initial design | .862 | .862 | – | .021 | .699 |
| | | | MedSigLIP LP | .917 | .917 | – | .031 | .527 |
| Middle East | HYGD | 548 | **LOTUS** | .909 | .909 | .909 | .104 | .715 |
| | | | Initial design | .850 | .850 | .850 | .068 | .689 |
| | | | MedSigLIP LP | .898 | .898 | .898 | .061 | .548 |
| South Asia | AFIO, Bangladesh Eye, IDRiD | 4,145 | **LOTUS** | .751 | .714 | .563 | .046 | .691 |
| | | | Initial design | .729 | .687 | .438 | .075 | .638 |
| | | | MedSigLIP LP | .678 | .702 | .443 | .087 | .601 |
| Mean | | | **LOTUS** | .822 | .800 | – | .061 | – |
| Mean | | | Initial design | .791 | .768 | – | .054 | – |
| Mean | | | MedSigLIP LP | .808 | .800 | – | .058 | – |

**Table S31.** Pairwise AUROC, mean over published test sources holding both classes, with bootstrap intervals over sources.

| Pair | A | B | Sources | AUROC A | AUROC B | A−B | 95% CI |
|---|---|---|---|---|---|---|---|
| D–G | 16-shot probe | Fine-tune | 3 | .939 | .916 | +0.023 | [-0.029, +0.077] |
| D–G | 16-shot probe | Linear probe | 3 | .939 | .917 | +0.022 | [-0.012, +0.059] |
| D–G | 16-shot probe | Group DRO | 3 | .939 | .942 | -0.003 | [-0.035, +0.033] |
| D–G | Linear probe | Fine-tune | 3 | .917 | .916 | +0.001 | [-0.062, +0.081] |
| D–G | Group DRO | ERM | 3 | .942 | .916 | +0.025 | [+0.010, +0.041] |
| N–D | 16-shot probe | Fine-tune | 4 | .946 | .900 | +0.046 | [+0.016, +0.074] |
| N–D | 16-shot probe | Linear probe | 4 | .946 | .928 | +0.019 | [-0.005, +0.053] |
| N–D | 16-shot probe | Group DRO | 4 | .946 | .923 | +0.024 | [+0.007, +0.045] |
| N–D | Linear probe | Fine-tune | 4 | .928 | .900 | +0.028 | [-0.012, +0.069] |
| N–D | Group DRO | ERM | 4 | .923 | .909 | +0.013 | [-0.021, +0.054] |
| N–G | 16-shot probe | Fine-tune | 4 | .749 | .643 | +0.106 | [+0.038, +0.203] |
| N–G | 16-shot probe | Linear probe | 4 | .749 | .678 | +0.071 | [-0.011, +0.152] |
| N–G | 16-shot probe | Group DRO | 4 | .749 | .704 | +0.045 | [-0.006, +0.118] |
| N–G | Linear probe | Fine-tune | 4 | .678 | .643 | +0.035 | [-0.059, +0.135] |
| N–G | Group DRO | ERM | 4 | .704 | .644 | +0.060 | [-0.015, +0.123] |

**Table S32.** Trained on development sources against trained within the target (logistic regression on frozen features, mean over nine encoders; within-source by cross-validation). Gap: within minus development bAcc, with its bootstrap interval.

| | bAcc | | ECE | | AUROC N–G | | AUROC N–D | | |
|---|---|---|---|---|---|---|---|---|---|
| Source | Dev | Within | Dev (TS) | Within | Dev | Within | Dev | Within | Gap |
| AFIO | .516 | .635 | .344 | .153 | .603 | .745 | .856 | .829 | +0.126 [-0.001, +0.249] |
| Chákṣu | .352 | .718 | .261 | .044 | .638 | .806 | – | – | +0.372 [+0.328, +0.415] |
| FIVES | .581 | .857 | .227 | .049 | .780 | .961 | .883 | .969 | +0.283 [+0.245, +0.320] |
| HRF | .583 | .779 | .295 | .182 | .477 | .854 | .926 | .927 | +0.193 [+0.053, +0.328] |
| PAPILA | .394 | .757 | .212 | .070 | .635 | .832 | – | – | +0.371 [+0.296, +0.445] |
| Paraguay | .817 | .928 | .046 | .018 | – | – | .934 | .975 | +0.115 [+0.095, +0.137] |

**Table S33.** More training sources at a fixed size (300 images per class): test and Chákṣu bAcc for $K = 1, 2, 4, 7$ development sources (mean over random pools).

| | Test | | | | Chákṣu | | | |
|---|---|---|---|---|---|---|---|---|
| Encoder | 1 | 2 | 4 | 7 | 1 | 2 | 4 | 7 |
| clip_b16 | .509 | .478 | .508 | .558 | .527 | .446 | .431 | .494 |
| convnext_t | .481 | .514 | .518 | .557 | .514 | .491 | .408 | .440 |
| dinov2_b14 | .544 | .618 | .593 | .720 | .535 | .555 | .485 | .543 |
| flair$^{\dagger}$ | .555 | .633 | .678 | .758 | .511 | .624 | .643 | .679 |
| medsiglip | .594 | .725 | .743 | .817 | .534 | .561 | .504 | .350 |
| resnet50 | .547 | .544 | .503 | .628 | .564 | .552 | .362 | .455 |
| siglip2_l16 | .512 | .518 | .548 | .602 | .494 | .456 | .387 | .102 |
| vit_b16_in21k | .417 | .478 | .542 | .576 | .504 | .456 | .347 | .430 |

**Table S34.** Learning curve: change in test bAcc at $k$ labels per class against the next smaller budget, with bootstrap intervals.

| Encoder | $k$ | ΔbAcc | 95% CI |
|---|---|---|---|
| medsiglip | 16 | -0.008 | [-0.161, +0.130] |
| medsiglip | 64 | +0.021 | [-0.111, +0.124] |
| medsiglip | 128 | +0.040 | [-0.060, +0.120] |
| medsiglip | 512 | +0.009 | [-0.054, +0.078] |
| dinov2_b14 | 16 | -0.130 | [-0.257, +0.032] |
| dinov2_b14 | 64 | -0.044 | [-0.151, +0.088] |
| dinov2_b14 | 128 | -0.016 | [-0.114, +0.089] |
| dinov2_b14 | 512 | +0.003 | [-0.076, +0.060] |
| siglip2_l16 | 16 | -0.050 | [-0.129, +0.024] |
| siglip2_l16 | 64 | -0.015 | [-0.096, +0.087] |
| siglip2_l16 | 128 | +0.025 | [-0.080, +0.125] |
| siglip2_l16 | 512 | +0.007 | [-0.103, +0.056] |

## S7 Attempted Fixes and Other Probes

**Table S35.** Test-time adaptation per held-out source: test bAcc for none, BatchNorm statistics, TENT and T3A ($K$ = 20, 100); Chákṣu bAcc for none, TENT and T3A-100.

| Setting | Model | None | BN | TENT | T3A-20 | T3A-100 | Ch. none | Ch. TENT | Ch. T3A |
|---|---|---|---|---|---|---|---|---|---|
| Fine-tune | swin_t | .655 | – | .655 | .666 | .668 | .298 | .297 | .309 |
| Fine-tune | resnet50 | .661 | .598 | .598 | .632 | .641 | .417 | .363 | .306 |
| Group DRO | convnext_t | .707 | – | .709 | .705 | .703 | .417 | .430 | .358 |
| ERM | dinov2_lora | .708 | – | .710 | .644 | .662 | .420 | .421 | .298 |
| 16-shot probe | medsiglip | .748 | – | – | .659 | .683 | .675 | – | .496 |
| Linear probe | medsiglip | .728 | – | – | .679 | .679 | .347 | – | .271 |

**Table S36.** Prior correction on top of TS: test bAcc with EM (Saerens et al.) pooled or per source, and with true priors pooled or per source (oracle, uses test labels); ECE for TS, EM per source and oracle per source.

| Setting | Model | TS | EM pooled | EM per src. | True pooled | Oracle per src. | ECE TS | ECE EM | ECE oracle |
|---|---|---|---|---|---|---|---|---|---|
| Zero-shot | medsiglip | .709 | .540 | .635 | .709 | .863 | .032 | .287 | .033 |
| CoOp | siglip_b16 | .491 | .398 | .457 | .513 | .737 | .175 | .433 | .050 |
| CoOp ctx-init | biomedclip | .497 | .416 | .438 | .495 | .754 | .150 | .389 | .040 |
| CoOp + LP | siglip_b16 | .496 | .394 | .450 | .520 | .736 | .176 | .445 | .053 |
| 16-shot probe | medsiglip | .748 | .683 | .787 | .769 | .853 | .044 | .185 | .051 |
| Linear probe | medsiglip | .728 | .778 | .835 | .778 | .853 | .079 | .120 | .081 |
| MLP probe | dinov2_b14 | .598 | .610 | .618 | .621 | .821 | .172 | .226 | .070 |
| Tip-Adapter-F (acc. rule) | biomedclip | .495 | .407 | .434 | .511 | .739 | .147 | .389 | .047 |
| Tip-Adapter (acc. rule) | biomedclip | .478 | .333 | .495 | .460 | .779 | .091 | .292 | .120 |
| Tip-Adapter-F | biomedclip | .495 | .407 | .434 | .511 | .739 | .147 | .389 | .047 |
| Tip-Adapter | biomedclip | .478 | .333 | .495 | .460 | .779 | .091 | .292 | .120 |
| Fine-tune | swin_t | .655 | .657 | .661 | .675 | .827 | .171 | .226 | .099 |
| Fine-tune, NLL rule | convnext_t | .677 | .687 | .710 | .707 | .842 | .133 | .182 | .073 |
| CORAL | convnext_t | .676 | .687 | .704 | .704 | .851 | .136 | .172 | .069 |
| SWAD | convnext_t | .672 | .691 | .704 | .706 | .851 | .139 | .166 | .060 |
| Group DRO | convnext_t | .707 | .720 | .723 | .733 | .857 | .091 | .188 | .051 |
| ERM | dinov2_lora | .707 | .655 | .726 | .727 | .826 | .115 | .218 | .104 |
| WiSE-FT | siglip2_l16 | .634 | .598 | .627 | .669 | .813 | .168 | .281 | .103 |

*Reading.* No label-free fix helps reliably. TENT changes little, BatchNorm adaptation harms ResNet-50 (notably on single-class PALM), and T3A is mostly negative. EM prior estimation is unstable and usually worsens ECE. True per-source priors raise bAcc to 0.81–0.86 for most settings, so class balance is a large part of the shift, but this oracle uses test labels, and single-class sources make it easy.

Table S37: $K$ labelled images from each target source, read on the rest of that source: ECE with the frozen $T$ against $T$ re-fitted on $K$; LAC .90 coverage with the frozen quantile against a quantile from $K$, with the 5th percentile over draws.

| Setting | Model | $K$ | ECE frozen | ECE $T$ from $K$ | LAC frozen | LAC from $K$ | LAC 5% |
|---|---|---|---|---|---|---|---|
| 16-shot probe | medsiglip | 10 | .156 | .128 | .879 | .897 | .714 |
| | | 25 | .152 | .093 | .869 | .920 | .801 |

Table S37: (continued)

| Setting | Model | $K$ | ECE frozen | ECE $T$ from $K$ | LAC frozen | LAC from $K$ | LAC 5% |
|---|---|---|---|---|---|---|---|
| | | 50 | .154 | .087 | .869 | .899 | .808 |
| | | 100 | .129 | .067 | .900 | .901 | .837 |
| Linear probe | medsiglip | 10 | .172 | .134 | .677 | .899 | .704 |
| | | 25 | .153 | .098 | .682 | .917 | .785 |
| | | 50 | .154 | .086 | .682 | .898 | .797 |
| | | 100 | .134 | .073 | .672 | .901 | .841 |
| Zero-shot | medsiglip | 10 | .163 | .136 | .884 | .901 | .722 |
| | | 25 | .155 | .103 | .884 | .920 | .809 |
| | | 50 | .156 | .094 | .884 | .900 | .813 |
| | | 100 | .121 | .080 | .919 | .902 | .843 |
| ERM | dinov2_lora | 10 | .200 | .140 | .658 | .904 | .714 |
| | | 25 | .189 | .110 | .663 | .920 | .819 |
| | | 50 | .191 | .104 | .663 | .899 | .805 |
| | | 100 | .194 | .089 | .643 | .901 | .831 |
| Group DRO | convnext_t | 10 | .189 | .140 | .712 | .903 | .715 |
| | | 25 | .178 | .117 | .721 | .916 | .803 |
| | | 50 | .180 | .109 | .721 | .899 | .800 |
| | | 100 | .160 | .101 | .718 | .901 | .837 |
| Fine-tune | swin_t | 10 | .283 | .125 | .590 | .904 | .712 |
| | | 25 | .276 | .095 | .588 | .921 | .809 |
| | | 50 | .277 | .088 | .588 | .901 | .811 |
| | | 100 | .285 | .066 | .557 | .902 | .838 |

*Reading.* Ten labels from the new source already bring LAC coverage near 0.90 on average and re-fitted $T$ lowers ECE, but the worst draws stay far below nominal at small $K$; this is a deployment recipe, not a guarantee.

*Reading.* CLAHE does not help fine-tuning or the 16-shot probe. The MedSigLIP linear probe rises on test, but its validation falls, so validation-based selection would not adopt it.

*Reading.* MSP accepts a third to two thirds of out-of-scope images. KNN rejects most of them and lowers risk, but largely by dropping whole hard sources; it is useful for flagging strange images rather than as a remedy for shift.

*Reading.* Generative multimodal models rarely call glaucoma. Mild DR (grade 1) is the hardest D grade for most models. Strong blur, and for most models JPEG compression, mainly remove glaucoma recall, consistent with the disc being a small, detail-dependent region; the 16-shot MedSigLIP probe is the exception under JPEG (0.784 to 0.870).

## S8 LOTUS: Specification and Development

### S8.1 Design search on development sources

The design search below produced the initial design; LOTUS removes its projection after the rotation ablation (Table S25). All results in this section are leave-one-development-source-out on development data, equal-site mean, three seeds unless noted, and use no test row. Table S45 compares losses (cross-entropy; a conditional-logit pair loss across sites, "cross"; the same within sites, "strata"), a single head against a leave-one-source-out bag, the between-site projection ("nap"), and text fusion with one bank or all three, on pooled MedSigLIP features.

**Table S38.** CLAHE against unprocessed images, same seeds and hardware.

| Setting | Config | Preprocessing | Val | Test | Eq-src | Chákṣu | ECE TS | Rec. G |
|---|---|---|---|---|---|---|---|---|
| Fine-tune | convnext_t_ce | none | .886 | .668 | .612 | .273 | .133 | .385 |
| Fine-tune | convnext_t_ce | CLAHE | .876 | .653 | .604 | .234 | .132 | .293 |
| Fine-tune | swin_t_mbls | none | .893 | .655 | .600 | .298 | .171 | .411 |
| Fine-tune | swin_t_mbls | CLAHE | .882 | .650 | .608 | .283 | .152 | .351 |
| Linear probe | medsiglip_class_ls | none | .920 | .728 | .654 | .347 | .079 | .488 |
| Linear probe | medsiglip_class_ls | CLAHE | .911 | .788 | .702 | .400 | .041 | .663 |
| 16-shot probe | medsiglip_none_ce | none | .802 | .748 | .683 | .675 | .044 | .783 |
| 16-shot probe | medsiglip_none_ce | CLAHE | .790 | .747 | .693 | .678 | .033 | .804 |

**Table S39.** Referral rules at a threshold for 80% coverage on Cal-B: maximum softmax (MSP), KNN distance in feature space, or both. Open: share of the 861 out-of-scope images accepted.

| Setting | Model | Score | Test cov. | Test risk | Cal-B cov. | Open | G→N |
|---|---|---|---|---|---|---|---|
| Fine-tune | swin_t | MSP | .694 | .250 | .801 | .666 | .147 |
| Fine-tune | swin_t | KNN | .415 | .084 | .801 | .165 | .007 |
| Fine-tune | swin_t | MSP+KNN | .484 | .113 | .802 | .252 | .020 |
| Group DRO | convnext_t | MSP | .738 | .182 | .801 | .602 | .089 |
| Group DRO | convnext_t | KNN | .400 | .076 | .801 | .101 | .010 |
| Group DRO | convnext_t | MSP+KNN | .487 | .094 | .802 | .187 | .026 |
| ERM | dinov2_lora | MSP | .715 | .223 | .801 | .610 | .067 |
| ERM | dinov2_lora | KNN | .549 | .266 | .801 | .216 | .113 |
| ERM | dinov2_lora | MSP+KNN | .628 | .221 | .804 | .290 | .092 |
| Linear probe | medsiglip | MSP | .610 | .129 | .801 | .337 | .061 |
| Linear probe | medsiglip | KNN | .419 | .227 | .801 | .059 | .093 |
| Linear probe | medsiglip | MSP+KNN | .421 | .153 | .802 | .074 | .057 |
| 16-shot probe | medsiglip | MSP | .753 | .195 | .801 | .679 | .012 |
| 16-shot probe | medsiglip | KNN | .419 | .223 | .801 | .059 | .027 |
| 16-shot probe | medsiglip | MSP+KNN | .483 | .183 | .802 | .105 | .020 |

**Table S40.** Out-of-scope detection AUROC from penultimate features of trained models (SWAD rows omitted: its saved checkpoint holds pre-averaging weights).

| Setting | Model | MSP | MaxLogit | Energy | KNN | Mahalanobis |
|---|---|---|---|---|---|---|
| Fine-tune | swin_t | .549 | .567 | .572 | .664 | .754 |
| Group DRO | convnext_t | .618 | .610 | .606 | .745 | .791 |
| ERM | dinov2_lora | .613 | .610 | .610 | .753 | .754 |

**Table S41.** Multiple-choice answering by multimodal LLMs (mean over six option orders).

| Model | Test bAcc | Rec. N | Rec. D | Rec. G | Chákṣu bAcc | Chákṣu Rec. G |
|---|---|---|---|---|---|---|
| Fundus-R1-7B | .576 | .789 | .795 | .143 | .463 | .128 |
| Qwen2.5-VL-7B-Instruct | .424 | .926 | .347 | .000 | .488 | .005 |
| llava-med-v1.5-mistral-7b-hf | .347 | .905 | .056 | .080 | .507 | .037 |
| llava-onevision-qwen2-7b-ov-hf | .407 | .367 | .855 | .000 | .096 | .000 |

**Table S42.** Recall of D by DR grade on test (grade −1: Davis grading without a numeric ICDR grade).

| Setting | Model | −1 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|---|
| Zero-shot | medsiglip | .785 | .000 | .734 | .954 | .838 |
| CoOp | siglip_b16 | .711 | .444 | .688 | .823 | .915 |
| CoOp ctx-init | biomedclip | .808 | .111 | .618 | .884 | .925 |
| CoOp + LP | siglip_b16 | .718 | .444 | .700 | .835 | .929 |
| 16-shot probe | medsiglip | .767 | .056 | .841 | .989 | .967 |
| Linear probe | medsiglip | .727 | .000 | .844 | .981 | .977 |
| MLP probe | dinov2_b14 | .665 | .222 | .737 | .935 | .932 |
| Tip-Adapter-F (acc. rule) | biomedclip | .660 | .167 | .569 | .831 | .885 |
| Tip-Adapter (acc. rule) | biomedclip | .368 | .000 | .122 | .432 | .499 |
| Tip-Adapter-F | biomedclip | .660 | .167 | .569 | .831 | .885 |
| Tip-Adapter | biomedclip | .368 | .000 | .122 | .432 | .499 |
| Fine-tune | swin_t | .724 | .389 | .844 | .974 | .986 |
| Fine-tune, NLL rule | convnext_t | .717 | .389 | .801 | .965 | .969 |
| CORAL | convnext_t | .704 | .278 | .801 | .966 | .972 |
| SWAD | convnext_t | .724 | .278 | .817 | .966 | .972 |
| Group DRO | convnext_t | .712 | .111 | .743 | .969 | .952 |
| ERM | dinov2_lora | .701 | .333 | .865 | .980 | .996 |
| WiSE-FT | siglip2_l16 | .695 | .222 | .755 | .964 | .968 |

**Table S43.** Glaucoma recall on clean test images and at the strongest blur and JPEG levels.

| Setting | Model | Clean | Blur | JPEG |
|---|---|---|---|---|
| Fine-tune | swin_t | .419 | .071 | .100 |
| Group DRO | convnext_t | .447 | .026 | .073 |
| ERM | dinov2_lora | .598 | .438 | .358 |
| 16-shot probe | medsiglip | .784 | .267 | .870 |
| Linear probe | medsiglip | .491 | .144 | .086 |

**Table S44.** Complete specification of LOTUS. The initial design adds the projection row before the heads.

| | |
|---|---|
| Encoder | MedSigLIP (patch 14, 448 px, 27 layers, width 1152), frozen |
| Head input | Last hidden state, $32 \times 32$ tokens pooled to $14 \times 14$; fixed random orthonormal map to 384 dimensions (seed 0); grid mean |
| Standardization | Per dimension, training-split mean and standard deviation |
| Projection | None. Initial design only: within-class between-site means, at most 32 directions, fitted per head on its training rows |
| Heads | One linear head per development source, trained without it; 1,155 parameters each |
| Training | Label smoothing 0.1; class-balanced sampling; AdamW, lr $3 \times 10^{-4}$, wd $10^{-2}$; batch 64; 12 epochs; best epoch by validation bAcc on the head's own sites |
| Rescaling | Centre per image; divide by the standard deviation over validation rows |
| Text | Mean of the three banks of Table S8, on the pooled image embedding |
| Fusion | 0.5 heads + 0.5 text; TS on Cal-A |
| Seeds | 0, 1, 2 on the published split and on every rotation; the main rotation table uses seed 0 for all methods |

Table S45: Factorial comparison on pooled features (development, three seeds).

| Arm | bAcc | ECE | NLL | Recall G |
|---|---|---|---|---|
| cross bag nap+allbanks | .794 | .156 | .574 | .636 |
| strata bag nap+allbanks | .792 | .151 | .573 | .625 |
| ce bag nap+allbanks | .791 | .155 | .581 | .639 |
| strata bag+allbanks | .784 | .148 | .584 | .618 |
| cross bag nap+text | .784 | .137 | .577 | .633 |
| cross bag+allbanks | .782 | .149 | .594 | .557 |
| strata single nap+allbanks | .782 | .144 | .578 | .692 |
| strata bag nap+text | .781 | .131 | .577 | .630 |
| ce bag nap+text | .780 | .134 | .584 | .640 |
| ce bag+allbanks | .777 | .149 | .604 | .576 |
| cross single nap+allbanks | .776 | .141 | .589 | .574 |
| strata bag+text | .774 | .129 | .589 | .622 |
| strata single nap+text | .773 | .128 | .583 | .696 |
| cross bag+text | .773 | .134 | .597 | .570 |
| cross single nap+text | .770 | .125 | .593 | .589 |
| strata single+allbanks | .770 | .135 | .595 | .594 |
| ce bag+text | .768 | .132 | .606 | .589 |
| ce single nap+allbanks | .765 | .136 | .601 | .583 |
| strata single+text | .762 | .120 | .599 | .607 |
| cross bag nap | .762 | .133 | .623 | .625 |
| cross single+allbanks | .760 | .135 | .616 | .472 |
| strata bag nap | .759 | .127 | .621 | .603 |
| ce single nap+text | .758 | .117 | .604 | .585 |
| cross single+text | .753 | .123 | .619 | .485 |
| ce single+allbanks | .752 | .143 | .622 | .531 |
| ce bag nap | .750 | .129 | .641 | .629 |
| strata bag | .747 | .130 | .640 | .597 |
| ce single+text | .746 | .130 | .624 | .541 |
| cross bag | .737 | .146 | .663 | .481 |
| strata single nap | .731 | .107 | .651 | .712 |
| cross single nap | .729 | .126 | .666 | .526 |
| strata single | .723 | .129 | .675 | .565 |
| ce bag | .721 | .145 | .687 | .507 |
| ce single nap | .704 | .125 | .694 | .517 |
| cross single | .691 | .158 | .715 | .319 |
| ce single | .675 | .169 | .741 | .417 |

*Reading*. Every component adds a little alone; with all three in place the loss choice no longer matters (0.791–0.794). Text with all banks is the largest single step.

Table S46: Head input (pooled feature, last-layer token mean, three depths, both) and head recipe (base: 40 epochs, lr $10^{-3}$, wd $10^{-4}$, batch 256; reg: 12 epochs, lr $3 \times 10^{-4}$, wd $10^{-2}$, batch 64), single head or bag with projection, with text.

| Study | Arm | bAcc | ECE | NLL | Recall G | False G |
|---|---|---|---|---|---|---|
| recipe | last reg bag nap+text | .792 | .151 | .575 | .764 | .100 |
| recipe | pooled base bag nap+text | .791 | .155 | .581 | .639 | .070 |
| input | pooled bag nap+text | .791 | .155 | .581 | .639 | .070 |
| recipe | last base bag nap+text | .787 | .150 | .582 | .678 | .078 |
| input | last bag nap+text | .787 | .150 | .582 | .678 | .078 |
| recipe | pooled reg bag nap+text | .787 | .155 | .580 | .783 | .109 |
| input | pooled last bag nap+text | .782 | .154 | .594 | .707 | .093 |
| recipe | pooled reg single+text | .759 | .136 | .621 | .688 | .109 |
| recipe | last reg single+text | .759 | .133 | .616 | .650 | .098 |
| recipe | last base single+text | .757 | .132 | .616 | .547 | .074 |
| input | last single+text | .757 | .132 | .616 | .547 | .074 |
| recipe | pooled base single+text | .752 | .143 | .622 | .531 | .071 |
| input | pooled single+text | .752 | .143 | .622 | .531 | .071 |
| input | pooled last single+text | .744 | .149 | .631 | .546 | .085 |
| input | depths bag nap+text | .731 | .118 | .642 | .613 | .118 |

Table S46: (continued)

| Study | Arm | bAcc | ECE | NLL | Recall G | False G |
|---|---|---|---|---|---|---|
| input | depths single+text | .696 | .139 | .678 | .520 | .126 |

**Table S47.** Initial design on each held-out development site.

| Held-out site | bAcc | Recall G | False G |
|---|---|---|---|
| Bangladesh Eye | .751 | .716 | .149 |
| DeepDRiD | .776 | – | .081 |
| GRAPE | .812 | .812 | – |
| IDRiD | .758 | – | .026 |
| Jichi | .767 | – | .101 |
| OIA-DDR | .730 | – | .202 |
| SUSTech-SYSU | .946 | – | .040 |

**Table S48.** Designs tried and not retained (development). Reference: pooled-feature pipeline 0.791; single cross-entropy head 0.675.

| Experiment | bAcc | Reading |
|---|---|---|
| Batch network predicting temperature and class bias (1 seed) | .539 | Below the raw head (.666) |
| EM prior shift with TS (1 seed) | .523 | Unreliable with few classes per source |
| Polar Fourier features around the disc | .702 | Below the pair loss alone (.723) |
| Site prior, $\kappa$ from 30 to 3000 | .693–.717 | No gain |
| Full residual corrector over logits | .667 | Recall G fell to 0 |
| Corrector with fusion weights only | .781 | ECE .057, bAcc did not rise |
| RPAttn head on three token depths (1 seed) | .664 | Recall G .008; with text .690 |
| RPAttn + MixStyle (1 seed) | .614 | Recall G .012; with text .688 |
| N/D from RPAttn, G from low-capacity branch (1 seed) | .788 | .793 for the pooled pipeline on that seed |
| Late fusion of nine encoder inputs | .765 | Other encoders lowered the score |
| Greedy nested encoder selection | .789 | Mostly kept MedSigLIP alone |
| Magnified optic-disc view with whole image | .768 | Recall G up (.767), false G up (.135) |
| Image-conditioned fusion gate | .763 | Below a plain average |
| Global class offset chosen on other sites | .679 | GRAPE fell to .006 |
| ConvNeXt-T end-to-end with pair loss (1 seed) | .622 | GRAPE .004 |

*Reading*. Heavier or learned modules repeatedly lost the glaucoma class at held-out sites, which led to the low-capacity design.

## S8.2 Error analysis and disc-side bias

*Reading*. On OIA-DDR, 1,629 of 5,639 normal images were predicted G although the source holds no glaucoma. Between 0.9% and 13.7% of a site's images were wrong under all nine inputs; among normal IDRiD images, 28%. A global class offset did not fix the G confusion, so it is a ranking problem.

**Table S49.** Held-out development sites for the pooled-feature pipeline: head, text and fusion bAcc, recalls, and images wrong under all nine encoder inputs.

| Held-out site | *n* | Head | Text | Fusion | N | D | G | All wrong |
|---|---|---|---|---|---|---|---|---|
| Bangladesh Eye | 3,153 | .651 | .732 | .753 | .761 | .963 | .536 | .039 |
| OIA-DDR | 11,269 | .660 | .738 | .735 | .685 | .786 | – | .137 |
| DeepDRiD | 1,804 | .803 | .721 | .795 | .682 | .908 | – | .111 |
| GRAPE | 528 | .644 | .750 | .741 | – | – | .741 | .104 |
| IDRiD | 462 | .757 | .752 | .751 | .527 | .974 | – | .100 |
| Jichi | 8,475 | .779 | .736 | .803 | .824 | .781 | – | .103 |
| SUSTech-SYSU | 1,097 | .975 | .889 | .957 | .938 | .975 | – | .009 |

**Table S50.** Normal images predicted G, by disc side (automatic, label-free disc location), held-out site.

| Held-out site | Disc left | Disc right |
|---|---|---|
| OIA-DDR | .386 | .191 |
| Bangladesh Eye | .141 | .023 |
| IDRiD | .119 | .045 |
| SUSTech-SYSU | .086 | .019 |
| Jichi | .067 | .054 |
| DeepDRiD | .096 | .086 |

Table S51: Upgrades of the initial design's input on development data: dimension (384 or 1152), mirrored training and averaging, NAP $k$, weight decay. A separate study with its own runs; its "d384 plain" baseline is the initial design's configuration but differs slightly from the main development ablation (0.789 against 0.792).

| Arm | bAcc | NLL | Recall G | False G | N→G left | N→G right |
|---|---|---|---|---|---|---|
| d384 mirror k32 wd0.1 | .792 | .577 | .658 | .078 | .127 | .127 |
| d384 mirror k16 wd0.1 | .792 | .577 | .658 | .078 | .127 | .127 |
| d384 mirror k32 wd0.01 | .792 | .577 | .652 | .076 | .125 | .124 |
| d384 mirror k16 wd0.01 | .792 | .577 | .652 | .076 | .125 | .124 |
| d384 mirror k32 wd0.001 | .792 | .578 | .653 | .077 | .124 | .124 |
| d384 mirror k16 wd0.001 | .792 | .578 | .653 | .077 | .124 | .124 |
| d384 mirror k8 wd0.001 | .790 | .578 | .650 | .077 | .126 | .125 |
| d384 mirror k8 wd0.01 | .790 | .579 | .648 | .077 | .126 | .124 |
| d384 mirror k8 wd0.1 | .790 | .578 | .644 | .077 | .125 | .124 |
| d384 plain k32 wd0.01 | .789 | .576 | .768 | .103 | .215 | .141 |
| d1152 mirror k32 wd0.01 | .778 | .603 | .711 | .100 | .176 | .165 |
| d1152 plain k32 wd0.01 | .770 | .603 | .823 | .134 | .296 | .188 |

*Reading.* Mirroring removes the left-right gap and lowers false G but costs recall G; all 1152 dimensions are worse; $k$ and weight decay hardly matter.

**Table S52.** Post-processing checked on held-out development sites, fitted on calibration rows of the other sites.

| State | ECE | NLL | Cov 80 cov. | Cov 80 risk | Risk 10 risk | LAC cov. | LAC size | Class-LAC | APS cov./size |
|---|---|---|---|---|---|---|---|---|---|
| Raw | .148 | .576 | .720 | .105 | .094 | .866 | 1.18 | .846 | .930/1.62 |
| TS | .052 | .502 | .719 | .104 | .096 | .867 | 1.18 | .842 | .946/1.78 |
| FTS | .053 | .505 | .720 | .106 | .096 | .867 | 1.18 | .843 | .919/1.61 |

### S8.3 Patch-level evidence

The head input is the mean of patch tokens, and standardization, projection and the linear head are all linear, so each head's logits split exactly into per-patch contributions: $z_s(x) = \frac{1}{P}\sum_p W_s(I - U_s U_s^\top)x_p + b_s$ for the initial design, and the same without the projection for LOTUS. The maps below were computed with the initial design. Figure S5 maps the glaucoma evidence (G logit minus the larger of N and D, averaged over heads) for the most and least confident glaucoma image of each glaucoma test source. Tokens recomputed for these images matched the stored ones to within 0.13 (feature scale about 90).

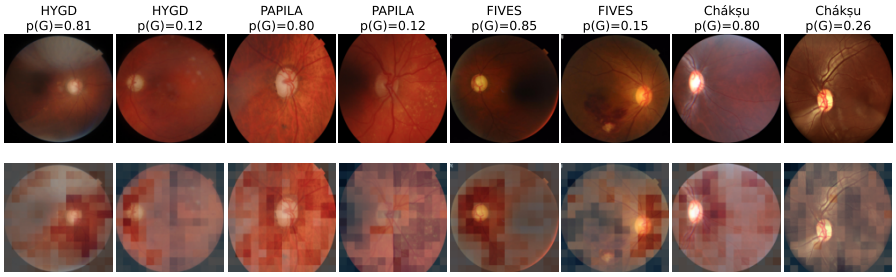



**Fig. S5.** Glaucoma evidence per patch (red: for G, blue: against), for the most and least confident glaucoma image of each source, $p(\mathrm{G})$ from raw fused logits. In confident cases the evidence concentrates on the optic disc; in missed cases it is weak and diffuse. Text scores are global and not mapped.

*Reading.* The maps are exact decompositions of the head, not post-hoc saliency, but they are not a validated localization of pathology.

### S8.4 Failure cases

*Reading.* Errors are often confident (0.82–1.00 after TS), which a global temperature cannot catch. Three patterns recur. Glaucoma missed as normal or DR occurs on sources with strong colour casts or low contrast (HRF, AFIO), where the disc is less distinct. Normal images called glaucoma show a prominent, bright disc. Normal or DR confusions on Paraguay and PALM involve visible bright or dark spots in images labelled normal, which may reflect label scope (Paraguay graded only DR) rather than model error; without re-grading we cannot tell. These observations are descriptive, not a clinical review.

### S8.5 Later attempts

Two further components were tried with selection rules written before their results were read; label scope was tested with the initial design, and the shift-aware reference is

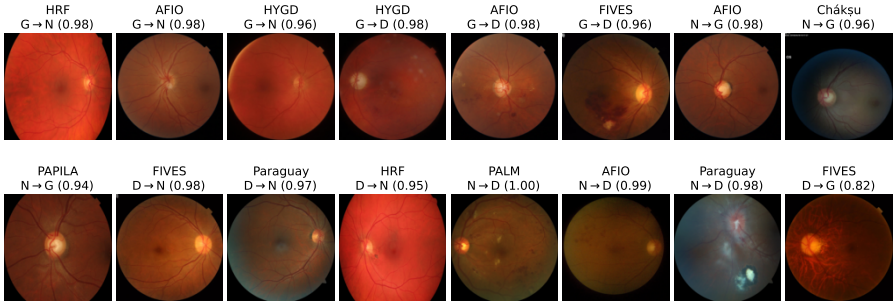



**Fig. S6.** Most confident errors of the initial design on test and Chákṣu (seed 0, after TS), up to three sources per error type: true→predicted class and confidence.

reported for LOTUS (Table S54; the initial design kept the risk target on 44 of 50 rotations against 23). *Label scope*: normal labels from DR-only sources were treated as the set {N, G}, with likelihood $p_{\mathrm{N}} + p_{\mathrm{G}}$. *Shift-aware reference*: temperature, thresholds and conformal quantiles were fitted on leave-one-source-out predictions rather than on Cal-B, with an optional normal certificate.

**Table S53.** Label scope, tested with the initial design: development (scope-aware bAcc counts a DR-only normal predicted G as correct) and the 50 rotations.

| Arm | bAcc | Scope-aware bAcc | ECE | Recall G | N→G, DR-only | N→G, full scope |
|---|---|---|---|---|---|---|
| exact | .792 | .844 | .151 | .764 | .145 | .322 |
| scope | .776 | .864 | .136 | .811 | .244 | .255 |
| exact (rotations) | .771 | .801 | .033 | .667 | – | .248 |
| scope (rotations) | .761 | .831 | .040 | .797 | – | .443 |

Table S54: LOTUS thresholds from Cal-B against the leave-one-source-out reference, 50 rotations, with and without the normal certificate; Held: rotations where test risk stayed at or below 10%. Here Cal-B is used whole; the main rotation table fits on one half of Cal-B and reads the other, so its LOTUS values differ in the third decimal.

| Fitted on | Target | Test cov. | Test risk | Held | FR | G→N | Chákṣu cov. | Chákṣu risk |
|---|---|---|---|---|---|---|---|---|
| Cal-B | cov. .80 | .724 | .102 | – | .069 | .062 | .383 | .293 |
| Cal-B | cov. .90 | .852 | .136 | – | .099 | .116 | .657 | .336 |
| Cal-B | risk .10 | .695 | .105 | 19/50 | .070 | .084 | .418 | .277 |
| Cal-B + cert. | cov. .80 | .596 | .096 | – | .037 | .024 | .232 | .437 |
| Cal-B + cert. | cov. .90 | .660 | .129 | – | .041 | .033 | .366 | .546 |
| Cal-B + cert. | risk .10 | .575 | .099 | 22/50 | .037 | .032 | .243 | .424 |
| LOSO reference | cov. .80 | .833 | .131 | – | .094 | .108 | .616 | .329 |
| LOSO reference | cov. .90 | .914 | .157 | – | .117 | .150 | .812 | .357 |
| LOSO reference | risk .10 | .592 | .075 | 42/50 | .046 | .038 | .192 | .252 |
| LOSO reference + cert. | cov. .80 | .665 | .121 | – | .038 | .028 | .343 | .514 |
| LOSO reference + cert. | cov. .90 | .705 | .146 | – | .038 | .028 | .437 | .585 |
| LOSO reference + cert. | risk .10 | .556 | .073 | 45/50 | .036 | .026 | .153 | .286 |

*Reading.* Label scope over-calls glaucoma at new sources. The shift-aware reference keeps coverage and risk promises far more often; risk at matched coverage is identical for both, so the arms differ only in where the threshold sits.

## S9 Statistical Comparisons

Table S55: Paired bootstrap over leakage groups on the published split, 2,000 draws, Holm within each metric; differences are LOTUS minus baseline. Adjusted $p = 0.009$ is the floor set by 2,000 draws after Holm correction. Intervals are conditional on these seven test sources and on LOTUS's test-based selection; they do not include source-level variance (bAcc SD 0.039 across rotations).

| Metric | Baseline | Baseline value | Diff. | 95% CI | Adj. $p$ |
|---|---|---|---|---|---|
| bAcc (raw) | 16-shot probe, MedSigLIP | .748 | +0.082 | [+0.067, +0.097] | 0.009 |
| bAcc (raw) | Linear probe, MedSigLIP | .728 | +0.102 | [+0.088, +0.116] | 0.009 |
| bAcc (raw) | Zero-shot, MedSigLIP | .709 | +0.122 | [+0.107, +0.136] | 0.009 |
| bAcc (raw) | ERM, DINOv2 LoRA | .707 | +0.123 | [+0.107, +0.138] | 0.009 |
| bAcc (raw) | Group DRO, ConvNeXt-T | .707 | +0.124 | [+0.109, +0.138] | 0.009 |
| bAcc (raw) | Fine-tune (NLL rule), ConvNeXt-T | .677 | +0.154 | [+0.136, +0.170] | 0.009 |
| bAcc (raw) | CORAL, ConvNeXt-T | .676 | +0.154 | [+0.137, +0.170] | 0.009 |
| bAcc (raw) | SWAD, ConvNeXt-T | .672 | +0.158 | [+0.141, +0.175] | 0.009 |
| bAcc (raw) | Fine-tune, Swin-T | .655 | +0.176 | [+0.157, +0.193] | 0.009 |
| bAcc (raw) | WiSE-FT, SigLIP2-L | .634 | +0.196 | [+0.177, +0.214] | 0.009 |
| bAcc (raw) | MLP probe, DINOv2-B | .598 | +0.232 | [+0.213, +0.250] | 0.009 |
| bAcc (raw) | CoOp ctx-init, BiomedCLIP | .497 | +0.334 | [+0.315, +0.352] | 0.009 |
| bAcc (raw) | CoOp + LP, SigLIP-B | .496 | +0.335 | [+0.316, +0.353] | 0.009 |
| bAcc (raw) | Tip-Adapter-F, BiomedCLIP | .495 | +0.336 | [+0.315, +0.356] | 0.009 |
| bAcc (raw) | CoOp, SigLIP-B | .491 | +0.339 | [+0.320, +0.357] | 0.009 |
| bAcc (raw) | Tip-Adapter, BiomedCLIP | .478 | +0.352 | [+0.332, +0.373] | 0.009 |
| ECE (TS) | Zero-shot, MedSigLIP | .032 | -0.002 | [-0.017, +0.013] | 0.799 |
| ECE (TS) | 16-shot probe, MedSigLIP | .044 | -0.014 | [-0.026, -0.000] | 0.091 |
| ECE (TS) | Linear probe, MedSigLIP | .079 | -0.050 | [-0.066, -0.028] | 0.009 |
| ECE (TS) | Group DRO, ConvNeXt-T | .091 | -0.062 | [-0.078, -0.043] | 0.009 |
| ECE (TS) | Tip-Adapter, BiomedCLIP | .091 | -0.062 | [-0.076, -0.044] | 0.009 |
| ECE (TS) | ERM, DINOv2 LoRA | .115 | -0.086 | [-0.103, -0.066] | 0.009 |
| ECE (TS) | Fine-tune (NLL rule), ConvNeXt-T | .133 | -0.104 | [-0.121, -0.081] | 0.009 |
| ECE (TS) | CORAL, ConvNeXt-T | .136 | -0.107 | [-0.124, -0.084] | 0.009 |
| ECE (TS) | SWAD, ConvNeXt-T | .139 | -0.109 | [-0.127, -0.087] | 0.009 |
| ECE (TS) | Tip-Adapter-F, BiomedCLIP | .147 | -0.118 | [-0.135, -0.100] | 0.009 |
| ECE (TS) | CoOp ctx-init, BiomedCLIP | .150 | -0.121 | [-0.140, -0.099] | 0.009 |
| ECE (TS) | WiSE-FT, SigLIP2-L | .168 | -0.139 | [-0.156, -0.117] | 0.009 |
| ECE (TS) | Fine-tune, Swin-T | .171 | -0.142 | [-0.158, -0.119] | 0.009 |
| ECE (TS) | MLP probe, DINOv2-B | .172 | -0.143 | [-0.160, -0.119] | 0.009 |
| ECE (TS) | CoOp, SigLIP-B | .175 | -0.146 | [-0.164, -0.124] | 0.009 |
| ECE (TS) | CoOp + LP, SigLIP-B | .176 | -0.147 | [-0.164, -0.124] | 0.009 |
| NLL (TS) | 16-shot probe, MedSigLIP | .644 | -0.187 | [-0.212, -0.162] | 0.009 |
| NLL (TS) | Zero-shot, MedSigLIP | .690 | -0.234 | [-0.254, -0.214] | 0.009 |
| NLL (TS) | Linear probe, MedSigLIP | .694 | -0.238 | [-0.271, -0.202] | 0.009 |
| NLL (TS) | Group DRO, ConvNeXt-T | .926 | -0.469 | [-0.536, -0.405] | 0.009 |
| NLL (TS) | ERM, DINOv2 LoRA | .944 | -0.487 | [-0.552, -0.430] | 0.009 |
| NLL (TS) | CORAL, ConvNeXt-T | 1.005 | -0.549 | [-0.613, -0.487] | 0.009 |
| NLL (TS) | Tip-Adapter, BiomedCLIP | 1.020 | -0.563 | [-0.592, -0.533] | 0.009 |
| NLL (TS) | Fine-tune (NLL rule), ConvNeXt-T | 1.036 | -0.579 | [-0.650, -0.513] | 0.009 |
| NLL (TS) | SWAD, ConvNeXt-T | 1.056 | -0.599 | [-0.672, -0.529] | 0.009 |
| NLL (TS) | Tip-Adapter-F, BiomedCLIP | 1.099 | -0.643 | [-0.679, -0.608] | 0.009 |
| NLL (TS) | Fine-tune, Swin-T | 1.174 | -0.717 | [-0.790, -0.645] | 0.009 |
| NLL (TS) | CoOp + LP, SigLIP-B | 1.180 | -0.723 | [-0.777, -0.669] | 0.009 |
| NLL (TS) | CoOp, SigLIP-B | 1.180 | -0.724 | [-0.776, -0.671] | 0.009 |
| NLL (TS) | CoOp ctx-init, BiomedCLIP | 1.189 | -0.733 | [-0.789, -0.675] | 0.009 |
| NLL (TS) | WiSE-FT, SigLIP2-L | 1.243 | -0.787 | [-0.863, -0.711] | 0.009 |

Table S55: (continued)

| Metric | Baseline | Baseline value | Diff. | 95% CI | Adj. $p$ |
|---|---|---|---|---|---|
| NLL (TS) | MLP probe, DINOv2-B | 1.259 | -0.802 | [-0.897, -0.708] | 0.009 |

**Table S56.** Configurations with development bAcc of at least 0.791, all evaluated on the published split; the one with the highest test bAcc (bold) became the initial design, and LOTUS is this configuration without the projection. All ten exceed the strongest validation-selected representative (0.748).

| Configuration | Dev | Test bAcc | Eq-src | Recall G | NLL (TS) | Chákṣu |
|---|---|---|---|---|---|---|
| Mirror, weight decay $10^{-3}$ | .800 | .809 | .726 | .696 | .503 | .714 |
| Mirror, weight decay $10^{-2}$ | .799 | .808 | .727 | .708 | .502 | .726 |
| Mirror, weight decay $10^{-4}$ | .795 | .807 | .724 | .681 | .511 | .703 |
| Cross-site pair loss | .794 | .809 | .717 | .722 | .507 | .694 |
| Same-site pair loss | .792 | .791 | .696 | .670 | .524 | .713 |
| **Tokens, weight decay $10^{-2}$ (selected)** | .792 | .817 | .724 | .765 | .482 | .731 |
| Pooled feature, weight decay $10^{-4}$ | .791 | .799 | .706 | .687 | .520 | .684 |
| Pooled feature + tokens | .791 | .812 | .719 | .719 | .493 | .701 |
| Disc side canonicalised | .791 | .805 | .733 | .679 | .511 | .637 |
| Mirrored text only | .791 | .800 | .714 | .679 | .512 | .704 |

### S9.1 Follow-up: clean selection, input versus recipe, prior correction, screening

All on the 50 rotations (`experiments/extra_runs.py`, `results/extra`).

**Table S57.** Follow-up results, means over rotations. Nested: arm chosen per rotation by leave-one-source-out on development only (LOTUS 29, initial design 8, single head + text 7, other 6). EM: label-free prior correction per test source.

| Model | bAcc | ECE (TS) | bAcc after EM | ECE after EM |
|---|---|---|---|---|
| Nested selection (no held-out data) | 0.780 | 0.035 | – | – |
| LOTUS (tokens) | 0.789 | 0.034 | 0.732 | 0.210 |
| LOTUS on pooled features | 0.789 | 0.035 | 0.737 | 0.204 |
| LOTUS head recipe, pooled, no bag/text | 0.763 | 0.055 | 0.720 | 0.220 |
| Probe recipe on tokens | 0.752 | 0.077 | 0.755 | 0.176 |
| Linear probe (pooled) | 0.746 | 0.084 | 0.750 | 0.182 |

*Reading.* Selecting without held-out data costs 0.009 and keeps the lead over the probe (39 of 50) and the calibration gain (47 of 50). The recipe and text, not the token input, carry the gain. Prior correction helps no model reliably and ruins calibration.

**Screening thresholds.** Thresholds on $P(G)$ set on Cal-B for 90% specificity gave, over rotations, test specificity 0.837 and sensitivity 0.857 for LOTUS (probe: 0.789 and 0.860), but on Chákṣu specificity 0.336 at sensitivity 0.954 (probe: 0.622 and 0.637). On the published split, 95% specificity on Cal-B gave 0.882 on test with sensitivity 0.798 (probe 0.865 and 0.708). Specificity targets, like risk targets, do not hold at new sources and must be set locally.

## S10 Reproducibility

Code, saved result tables and this document are archived at `https://doi.org/10.5281/zenodo.23979471` (GitHub: `https://github.com/dangaura-tech/fundusshift`). The README maps each result to its script and table file. LOTUS and its ablation over the rotations and regions are produced by `experiments/revision_runs.py` (saved in `results/revision`); its published-split evaluation and paired tests by `experiments/lotus_eval.py`, its rotation promises by `experiments/lotus_rotation.py`, and its per-source and disease-level tables by `experiments/review_tables.py` (saved in `results/lotus`). The evaluation protocol, including the selection rules written before each later attempt, is `docs/protocol.md`. Images are obtained from their original publishers; split tables are available from the author on request.